with the following functional dependencies:

I. $\text{Title Author} \rightarrow \text{Catalog\_no}$
II. $\text{Catalog\_no} \rightarrow \text{Title Author Publisher Year}$
III. $\text{Publisher Title Year} \rightarrow \text{Price}$

Assume $\{\text{Author, Title}\}$ is the key for both schemes. Which of the following statements is true?

A. Both Book and Collection are in BCNF
B. Both Book and Collection are in 3NF only
C. Book is in 2NF and Collection in 3NF
D. Both Book and Collection are in 2NF only

gatecse-2008 databases database-normalization normal

Answer key

3.6.27 Database Normalization: GATE CSE 2012 | Question: 2

Which of the following is **TRUE**?

A. Every relation in 3NF is also in BCNF
B. A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent on every key of R
C. Every relation in BCNF is also in 3NF
D. No relation can be in both BCNF and 3NF

gatecse-2012 databases easy database-normalization

Answer key

3.6.28 Database Normalization: GATE CSE 2013 | Question: 54

Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values. $F = \{CH \rightarrow G, A \rightarrow BC, B \rightarrow CFH, E \rightarrow A, F \rightarrow EG\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R.

How many candidate keys does the relation R have?

A. 3
B. 4
C. 5
D. 6

gatecse-2013 databases database-normalization normal

Answer key

3.6.29 Database Normalization: GATE CSE 2013 | Question: 55

Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values. $F = \{CH \rightarrow G, A \rightarrow BC, B \rightarrow CFH, E \rightarrow A, F \rightarrow EG\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R.

The relation R is

A. in 1NF, but not in 2NF.
B. in 2NF, but not in 3NF.
C. in 3NF, but not in BCNF.
D. in BCNF.

gatecse-2013 databases database-normalization normal

Answer key

3.6.30 Database Normalization: GATE CSE 2014 | Set 1 | Question: 21

Consider the relation scheme $R = (E, F, G, H, I, J, K, L, M, N)$ and the set of functional dependencies

$$\{\{E, F\} \rightarrow \{G\}, \{F\} \rightarrow \{I, J\}, \{E, H\} \rightarrow \{K, L\},\\ \{K\} \rightarrow \{M\}, \{L\} \rightarrow \{N\}\}$$

on R. What is the key for R?

A. $\{E, F\}$
B. $\{E, F, H\}$
C. $\{E, F, H, K, L\}$
D. $\{E\}$

gatecse-2014-set1 databases database-normalization normal

Answer key

3.6.31 Database Normalization: GATE CSE 2014 | Set 1 | Question: 30

Given the following two statements:

S1: Every table with two single-valued attributes is in 1NF, 2NF, 3NF and BCNF.

S2: $AB \rightarrow C, D \rightarrow E, E \rightarrow C$ is a minimal cover for the set of functional dependencies $AB \rightarrow C, D \rightarrow E, AB \rightarrow E, E \rightarrow C$.

Which one of the following is **CORRECT**?

A. S1 is TRUE and S2 is FALSE.
B. Both S1 and S2 are TRUE.
C. S1 is FALSE and S2 is TRUE.
D. Both S1 and S2 are FALSE.

gatecse-2014-set1 databases database-normalization normal

Answer key

3.6.32 Database Normalization: GATE CSE 2015 | Set 3 | Question: 20

Consider the relation $X(P, Q, R, S, T, U)$ with the following set of functional dependencies

$$F = \{$$
$$\{P, R\} \rightarrow \{S, T\},$$
$$\{P, S, U\} \rightarrow \{Q, R\}$$
$$\}$$

Which of the following is the trivial functional dependency in F^+, where F^+ is closure to F?

A. $\{P,R\} \rightarrow \{S,T\}$
B. $\{P,R\} \rightarrow \{R,T\}$
C. $\{P,S\} \rightarrow \{S\}$
D. $\{P,S,U\} \rightarrow \{Q\}$

gatecse-2015-set3 databases database-normalization easy

Answer key

3.6.33 Database Normalization: GATE CSE 2016 | Set 1 | Question: 21

Which of the following is NOT a superkey in a relational schema with attributes V, W, X, Y, Z and primary key $V\ Y$?

A. $VXYZ$
B. $VWXZ$
C. $VWXY$
D. $VWXYZ$

gatecse-2016-set1 databases database-normalization easy

Answer key

3.6.34 Database Normalization: GATE CSE 2016 | Set 1 | Question: 23

A database of research articles in a journal uses the following schema.

$$(\text{VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, YEAR, PRICE})$$

The primary key is '$(\text{VOLUME, NUMBER, STARTPAGE, ENDPAGE})$

and the following functional dependencies exist in the schema.

$$(\text{VOLUME , NUMBER, STARTPAGE, ENDPAGE}) \rightarrow \text{TITLE}$$
$$(\text{VOLUME, NUMBER}) \rightarrow \text{YEAR}$$
$$(\text{VOLUME, NUMBER, STARTPAGE, ENDPAGE}) \rightarrow \text{PRICE}$$

The database is redesigned to use the following schemas

$$(\text{VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, PRICE})$$
$$(\text{VOLUME, NUMBER, YEAR})$$

Which is the weakest normal form that the new database satisfies, but the old one does not?

A. 1NF
B. 2NF
C. 3NF
D. BCNF

gatecse-2016-set1 databases database-normalization normal

Answer key☟

3.6.35 Database Normalization: GATE CSE 2017 | Set 1 | Question: 16

The following functional dependencies hold true for the relational schema $R\{V, W, X, Y, Z\}$:

- $V \rightarrow W$
- $VW \rightarrow X$
- $Y \rightarrow VX$
- $Y \rightarrow Z$

Which of the following is irreducible equivalent for this set of functional dependencies?

A. $V \rightarrow W$
$V \rightarrow X$
$Y \rightarrow V$
$Y \rightarrow Z$

B. $V \rightarrow W$
$W \rightarrow X$
$Y \rightarrow V$
$Y \rightarrow Z$

C. $V \rightarrow W$
$V \rightarrow X$
$Y \rightarrow V$
$Y \rightarrow X$
$Y \rightarrow Z$

D. $V \rightarrow W$
$W \rightarrow X$
$Y \rightarrow V$
$Y \rightarrow X$
$Y \rightarrow Z$

gatecse-2017-set1 databases database-normalization normal

Answer key☟

3.6.36 Database Normalization: GATE CSE 2018 | Question: 42

Consider the following four relational schemas. For each schema , all non-trivial functional dependencies are listed, The **bolded** attributes are the respective primary keys.

Schema I: Registration(rollno, courses)

Field 'courses' is a set-valued attribute containing the set of courses a student has registered for.

Non-trivial functional dependency

rollno $\rightarrow$ courses

Schema II: Registration (rollno, coursid, email)

Non-trivial functional dependencies:

rollno, courseid $\rightarrow$ email

email $\rightarrow$ rollno

Schema III: Registration (rollno, courseid, marks, grade)

Non-trivial functional dependencies:

rollno, courseid, $\rightarrow$ marks, grade

marks $\rightarrow$ grade

Schema IV: Registration (rollno, courseid, credit)

Non-trivial functional dependencies:

rollno, courseid $\rightarrow$ credit

courseid $\rightarrow$ credit

Which one of the relational schemas above is in 3NF but not in BCNF?

A. Schema I
B. Schema II
C. Schema III
D. Schema IV

gatecse-2018 databases database-normalization normal two-marks

Answer key☟

3.6.37 Database Normalization: GATE CSE 2019 | Question: 32

Let the set of functional dependencies $F = \{QR \rightarrow S,\ R \rightarrow P,\ S \rightarrow Q\}$ hold on a relation schema $X = (PQRS)$. X is not in BCNF. Suppose X is decomposed into two schemas Y and Z, where $Y = (PR)$ and $Z = (QRS)$.

Consider the two statements given below.

I. Both Y and Z are in BCNF
II. Decomposition of X into Y and Z is dependency preserving and lossless

Which of the above statements is/are correct?

A. Both I and II B. I only C. II only D. Neither I nor II

gatecse-2019 databases database-normalization two-marks

Answer key

3.6.38 Database Normalization: GATE CSE 2020 | Question: 36

Consider a relational table R that is in $3NF$, but not in BCNF. Which one of the following statements is TRUE?

A. R has a nontrivial functional dependency $X \to A$, where X is not a superkey and A is a prime attribute.
B. R has a nontrivial functional dependency $X \to A$, where X is not a superkey and A is a non-prime attribute and X is not a proper subset of any key.
C. R has a nontrivial functional dependency $X \to A$, where X is not a superkey and A is a non-prime attribute and X is a proper subset of some key
D. A cell in R holds a set instead of an atomic value.

gatecse-2020 databases database-normalization two-marks

Answer key

3.6.39 Database Normalization: GATE CSE 2021 | Set 1 | Question: 33

Consider the relation $R(P,Q,S,T,X,Y,Z,W)$ with the following functional dependencies.

$$PQ \to X; \quad P \to YX; \quad Q \to Y; \quad Y \to ZW$$

Consider the decomposition of the relation R into the constituent relations according to the following two decomposition schemes.

- $D_1: \quad R = [(P,Q,S,T);\ (P,T,X);\ (Q,Y);\ (Y,Z,W)]$
- $D_2: \quad R = [(P,Q,S);\ (T,X);\ (Q,Y);\ (Y,Z,W)]$

Which one of the following options is correct?

A. D_1 is a lossless decomposition, but D_2 is a lossy decomposition
B. D_1 is a lossy decomposition, but D_2 is a lossless decomposition
C. Both D_1 and D_2 are lossless decompositions
D. Both D_1 and D_2 are lossy decompositions

gatecse-2021-set1 databases database-normalization two-marks

Answer key

3.6.40 Database Normalization: GATE CSE 2021 | Set 2 | Question: 40

Suppose the following functional dependencies hold on a relation U with attributes P, Q, R, S, and T:

- $P \to QR$
- $RS \to T$

Which of the following functional dependencies can be inferred from the above functional dependencies?

A. $PS \to T$ B. $R \to T$ C. $P \to R$ D. $PS \to Q$

gatecse-2021-set2 multiple-selects databases database-normalization two-marks

Answer key

3.6.41 Database Normalization: GATE CSE 2022 | Question: 21

Consider a relation $R(A, B, C, D, E)$ with the following three functional dependencies.

$AB \rightarrow C;\ BC \rightarrow D;\ C \rightarrow E;$

The number of superkeys in the relation R is ______________ .

gatecse-2022 numerical-answers databases database-normalization one-mark

Answer key

3.6.42 Database Normalization: GATE CSE 2022 | Question: 4

In a relational data model, which one of the following statements is TRUE?

A. A relation with only two attributes is always in BCNF.
B. If all attributes of a relation are prime attributes, then the relation is in BCNF.
C. Every relation has at least one non-prime attribute.
D. BCNF decompositions preserve functional dependencies.

gatecse-2022 databases database-normalization one-mark easy

Answer key

3.6.43 Database Normalization: GATE CSE 2024 | Set 1 | Question: 12

Which of the following statements about a relation $\mathbf{R}$ in first normal form (1NF) is/are TRUE?

A. $\mathbf{R}$ can have a multi-attribute key
B. $\mathbf{R}$ cannot have a foreign key
C. $\mathbf{R}$ cannot have a composite attribute
D. $\mathbf{R}$ cannot have more than one candidate key

gatecse-2024-set1 multiple-selects databases database-normalization one-mark

Answer key

3.6.44 Database Normalization: GATE CSE 2024 | Set 1 | Question: 34

The symbol $\rightarrow$ indicates functional dependency in the context of a relational database. Which of the following options is/are TRUE?

A. $(X, Y) \rightarrow (Z, W)$ implies $X \rightarrow (Z, W)$
B. $(X, Y) \rightarrow (Z, W)$ implies $(X, Y) \rightarrow Z$
C. $((X, Y) \rightarrow Z$ and $W \rightarrow Y)$ implies $(X, W) \rightarrow Z$
D. $(X \rightarrow Y$ and $Y \rightarrow Z)$ implies $X \rightarrow Z$

gatecse-2024-set1 multiple-selects databases database-normalization two-marks

Answer key

3.6.45 Database Normalization: GATE CSE 2024 | Set 2 | Question: 46

A functional dependency $F : X \rightarrow Y$ is termed as a useful functional dependency if and only if it satisfies all the following three conditions:

- X is not the empty set.
- Y is not the empty set.
- Intersection of X and Y is the empty set.

For a relation R with 4 attributes, the total number of possible useful functional dependencies is __________.

gatecse-2024-set2 numerical-answers databases database-normalization two-marks

Answer key

3.6.46 Database Normalization: GATE CSE 2025 | Set 2 | Question: 36

Consider the following relational schema along with all the functional dependencies that hold on them.

$$R1(A, B, C, D, E) : \{D \rightarrow E, EA \rightarrow B, EB \rightarrow C\}$$
$$R2(A, B, C, D) : \{A \rightarrow D, A \rightarrow B, C \rightarrow A\}$$

Which of the following statement(s) is/are TRUE?

A. $R1$ is in 3 NF
B. $R2$ is in 3 NF
C. $R1$ is NOT in 3 NF
D. $R2$ is NOT in 3 NF

gatecse2025-set2 databases database-normalization multiple-selects two-marks

Answer key

3.6.47 Database Normalization: GATE CSE 2026 | Set 1 | Question: 21

In the context of relational database normalization, which of the following statements is/are true?

A. It is always possible to obtain a dependency-preserving 3NF decomposition of a relation
B. It is always possible to obtain a dependency-preserving 1NF decomposition of a relation
C. It is not always possible to obtain a dependency-preserving BCNF decomposition of a relation
D. It is not always possible to obtain a dependency-preserving 2NF decomposition of a relation

gatecse-2026-set1 databases database-normalization multiple-selects one-mark

Answer key

3.6.48 Database Normalization: GATE DS&AI 2024 | Question: 36

Given the relational schema $R = (U, V, W, X, Y, Z)$ and the set of functional dependencies:

$$\{U \rightarrow V, U \rightarrow W, WX \rightarrow Y, WX \rightarrow Z, V \rightarrow X\}$$

Which of the following functional dependencies can be derived from the above set?

A. $VW \rightarrow YZ$
B. $WX \rightarrow YZ$
C. $VW \rightarrow U$
D. $VW \rightarrow Y$

gate-ds-ai-2024 databases database-normalization multiple-selects two-marks

Answer key

3.6.49 Database Normalization: GATE Data Science and Artificial Intelligence 2024 | Sample Paper | Question: 26

Given the following relation instances

X	Y	Z
1	4	2
1	5	3
1	4	3
1	5	2
3	2	1

Which of the following conditions is/are **TRUE**?

A. $\mathrm{XY} -> \mathrm{Z}$ and $\mathrm{Z} -> \mathrm{Y}$

B. $\mathrm{YZ}- > \mathrm{X}$ and $\mathrm{X}- >> \mathrm{Y}$
C. $\mathrm{Y}- > \mathrm{X}$ and $\mathrm{Y}- >> \mathrm{X}$
D. $\mathrm{XZ}- > \mathrm{Y}$ and $\mathrm{Y}- > \mathrm{X}$

gateda-sample-paper-2024 database-normalization

Answer key

3.6.50 Database Normalization: GATE IT 2004 | Question: 75

A relation Empdtl is defined with attributes empcode (unique), name, street, city, state and pincode. For any pincode, there is only one city and state. Also, for any given street, city and state, there is just one pincode. In normalization terms, Empdtl is a relation in

A. 1NF only
B. 2NF and hence also in 1NF
C. 3NF and hence also in 2NF and 1NF
D. BCNF and hence also in 3NF, 2NF and 1NF

gateit-2004 databases database-normalization normal

Answer key

3.6.51 Database Normalization: GATE IT 2005 | Question: 22

A table has fields F_1, F_2, F_3, F_4, F_5 with the following functional dependencies

- $F_1 \to F_3, F_2 \to F_4, (F_1 . F_2) \to F_5$

In terms of Normalization, this table is in

A. 1 NF
B. 2 NF
C. 3 NF
D. None of these

gateit-2005 databases database-normalization easy

Answer key

3.6.52 Database Normalization: GATE IT 2005 | Question: 70

In a schema with attributes A, B, C, D and E following set of functional dependencies are given

- $A \to B$
- $A \to C$
- $CD \to E$
- $B \to D$
- $E \to A$

Which of the following functional dependencies is NOT implied by the above set?

A. $CD \to AC$
B. $BD \to CD$
C. $BC \to CD$
D. $AC \to BC$

gateit-2005 databases database-normalization normal

Answer key

3.6.53 Database Normalization: GATE IT 2006 | Question: 60

Consider a relation R with five attributes $V, W, X, Y,$ and $Z.$ The following functional dependencies hold: $VY \to W, WX \to Z,$ and $ZY \to V.$
Which of the following is a candidate key for R?

A. VXZ
B. VXY
C. $VWXY$
D. $VWXYZ$

gateit-2006 databases database-normalization normal

Answer key

3.6.54 Database Normalization: GATE IT 2008 | Question: 61

Let $R(A, B, C, D)$ be a relational schema with the following functional dependencies : $A \to B$, $B \to C$, $C \to D$ and $D \to B$. The decomposition of R into $(A, B), (B, C), (B, D)$

A. gives a lossless join, and is dependency preserving
B. gives a lossless join, but is not dependency preserving
C. does not give a lossless join, but is dependency preserving
D. does not give a lossless join and is not dependency preserving

gateit-2008 databases database-normalization normal

Answer key

3.6.55 Database Normalization: GATE IT 2008 | Question: 62

Let $R(A, B, C, D, E, P, G)$ be a relational schema in which the following functional dependencies are known to hold: $AB \to CD, DE \to P, C \to E, P \to C$ and $B \to G$. The relational schema R is

A. in BCNF
B. in 3NF, but not in BCNF
C. in 2NF, but not in 3NF
D. not in 2NF

gateit-2008 databases database-normalization normal

Answer key

3.6.56 Database Normalization: GATE2001-1.23, UGCNET-June2012-III: 18

Consider a schema $R(A, B, C, D)$ and functional dependencies $A \to B$ and $C \to D$. Then the decomposition of R into $R_1(A, B)$ and $R_2(C, D)$ is

A. dependency preserving and lossless join
B. lossless join but not dependency preserving
C. dependency preserving but not lossless join
D. not dependency preserving and not lossless join

gate1998 databases ugcnetcse-june2012-paper3 database-normalization

Answer key

3.7 Database Schema (1)

3.7.1 Database Schema: GATE CSE 2026 | Set 2 | Question: 5

In the context of DBMS, consider the two sets $\mathbf{T}$ and $\mathbf{S}$ given below.

T	S
I: Logical schema	L: Views
II: Physical schema	M: File organization and indexes
III: External schema	N: Relations

Which one of the following is the correct match from $\mathbf{T}$ to $\mathbf{S}$?

A. $\text{I} - \text{L}, \text{II} - \text{M}, \text{III} - \text{N}$
B. $\text{I} - \text{M}, \text{II} - \text{L}, \text{III} - \text{N}$
C. $\text{I} - \text{N}, \text{II} - \text{M}, \text{III} - \text{L}$
D. $\text{I} - \text{N}, \text{II} - \text{L}, \text{III} - \text{M}$

gatecse-2026-set2 databases database-schema easy one-mark

Answer key

3.8 Decomposition (1)

3.8.1 Decomposition: GATE DA 2025 | Question: 6

If a relational decomposition is not dependency-preserving, which one of the following relational operators will be executed more frequently in order to maintain the dependencies?

A. Selection B. Projection C. Join D. Set union

gateda-2025 databases decomposition relational-algebra easy one-mark

Answer key

3.9 ER Diagram (12)

Practice Tests: Test 1 (15Q) Test 2 (6Q)

3.9.1 ER Diagram: GATE CSE 2005 | Question: 75

Let E_1 and E_2 be two entities in an E/R diagram with simple-valued attributes. R_1 and R_2 are two relationships between E_1 and E_2, where R_1 is one-to-many and R_2 is many-to-many. R_1 and R_2 do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model?

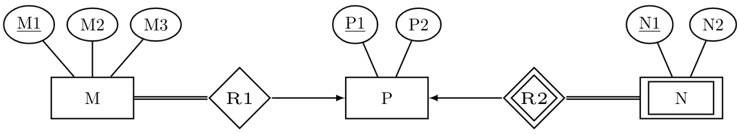
A. 2 B. 3 C. 4 D. 5

gatecse-2005 databases er-diagram normal

Answer key

3.9.2 ER Diagram: GATE CSE 2008 | Question: 82

Consider the following ER diagram



The minimum number of tables needed to represent M, N, P, $R1$, $R2$ is

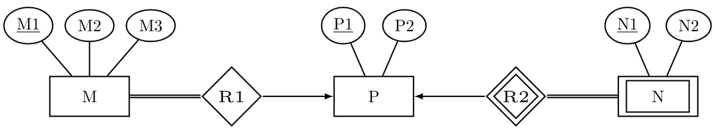
A. 2 B. 3 C. 4 D. 5

gatecse-2008 databases er-diagram normal

Answer key

3.9.3 ER Diagram: GATE CSE 2008 | Question: 83

Consider the following ER diagram



The minimum number of tables needed to represent M, N, P, $R1$, $R2$ is

Which of the following is a correct attribute set for one of the tables for the minimum number of tables needed to represent M, N, P, $R1$, $R2$?

A. $M1, M2, M3, P1$ B. $M1, P1, N1, N2$ C. $M1, P1, N1$ D. $M1, P1$

gatecse-2008 databases er-diagram normal

Answer key

3.9.4 ER Diagram: GATE CSE 2012 | Question: 14

Given the basic ER and relational models, which of the following is **INCORRECT**?

A. An attribute of an entity can have more than one value

B. An attribute of an entity can be composite
C. In a row of a relational table, an attribute can have more than one value
D. In a row of a relational table, an attribute can have exactly one value or a NULL value

gatecse-2012 databases normal er-diagram

Answer key

3.9.5 ER Diagram: GATE CSE 2015 | Set 1 | Question: 41

Consider an Entity-Relationship (ER) model in which entity sets E_1 and E_2 are connected by an $m : n$ relationship R_{12}. E_1 and E_3 are connected by a $1 : n$ $(1$ on the side of E_1 and n on the side of $E_3)$ relationship R_{13}.

E_1 has two-singled attributes a_{11} and a_{12} of which a_{11} is the key attribute. E_2 has two singled-valued attributes a_{21} and a_{22} of which a_{21} is the key attribute. E_3 has two single-valued attributes a_{31} and a_{32} of which a_{31} is the key attribute. The relationships do not have any attributes.

If a relational model is derived from the above ER model, then the minimum number of relations that would be generated if all relation are in $\mathrm{3NF}$ is________________.

gatecse-2015-set1 databases er-diagram normal numerical-answers

Answer key

3.9.6 ER Diagram: GATE CSE 2017 | Set 2 | Question: 17

An ER model of a database consists of entity types A and B. These are connected by a relationship R which does not have its own attribute. Under which one of the following conditions, can the relational table for R be merged with that of A?

A. Relationship R is one-to-many and the participation of A in R is total
B. Relationship R is one-to-many and the participation of A in R is partial
C. Relationship R is many-to-one and the participation of A in R is total
D. Relationship R is many-to-one and the participation of A in R is partial

gatecse-2017-set2 databases er-diagram normal

Answer key

3.9.7 ER Diagram: GATE CSE 2018 | Question: 11

In an Entity-Relationship (ER) model, suppose R is a many-to-one relationship from entity set E1 to entity set E2. Assume that E1 and E2 participate totally in R and that the cardinality of E1 is greater than the cardinality of E2.

Which one of the following is true about R?

A. Every entity in E1 is associated with exactly one entity in E2
B. Some entity in E1 is associated with more than one entity in E2
C. Every entity in E2 is associated with exactly one entity in E1
D. Every entity in E2 is associated with at most one entity in E1

gatecse-2018 databases er-diagram normal one-mark

Answer key

3.9.8 ER Diagram: GATE CSE 2020 | Question: 14

Which one of the following is used to represent the supporting many-one relationships of a weak entity set in an entity-relationship diagram?

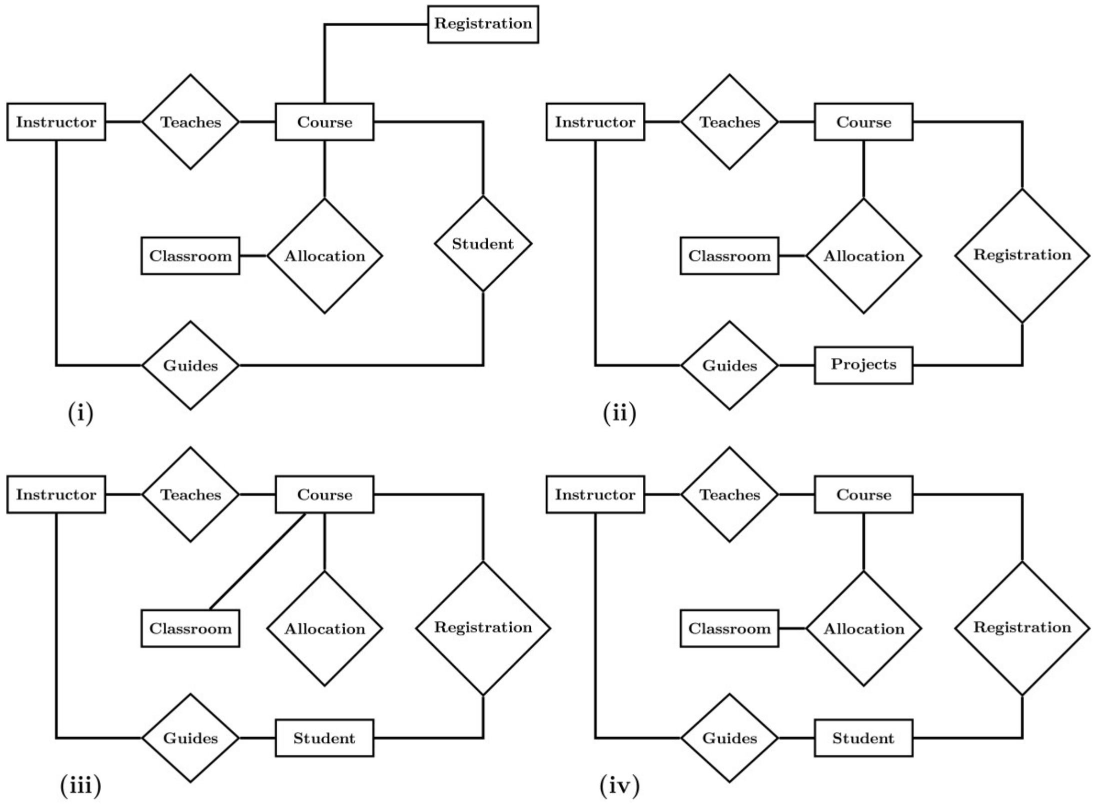
A. Diamonds with double/bold border
B. Rectangles with double/bold border
C. Ovals with double/bold border
D. Ovals that contain underlined identifiers

gatecse-2020 databases er-diagram one-mark

Answer key

3.9.9 ER Diagram: GATE CSE 2024 | Set 1 | Question: 10

Let S be the specification: "Instructors teach courses. Students register for courses. Courses are allocated classrooms. Instructors guide students." Which one of the following ER diagrams CORRECTLY represents S?



A. (i)
B. (ii)
C. (iii)
D. (iv)

gatecse-2024-set1 databases er-diagram one-mark

Answer key

3.9.10 ER Diagram: GATE CSE 2024 | Set 2 | Question: 10

In the context of owner and weak entity sets in the ER (Entity-Relationship) data model, which one of the following statements is TRUE?

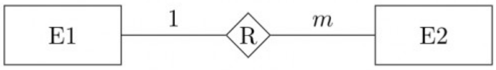
A. The weak entity set MUST have total participation in the identifying relationship
B. The owner entity set MUST have total participation in the identifying relationship
C. Both weak and owner entity sets MUST have total participation in the identifying relationship
D. Neither weak entity set nor owner entity set MUST have total participation in the identifying relationship

gatecse-2024-set2 databases er-diagram one-mark

Answer key

3.9.11 ER Diagram: GATE IT 2004 | Question: 73

Consider the following entity relationship diagram (ERD), where two entities $E1$ and $E2$ have a relation R of cardinality 1:m.



The attributes of $E1$ are $A11$, $A12$ and $A13$ where $A11$ is the key attribute. The attributes of $E2$ are $A21$, $A22$ and $A23$ where $A21$ is the key attribute and $A23$ is a multi-valued attribute. Relation R does not have any attribute. A relational database containing minimum number of tables with each table satisfying the requirements of the third normal form $(3NF)$ is designed from the above ERD. The number of tables in the database is

A. 2
B. 3
C. 5
D. 4

gateit-2004 databases er-diagram normal

Answer key☟

3.9.12 ER Diagram: GATE IT 2005 | Question: 21

Consider the entities 'hotel room', and 'person' with a many to many relationship 'lodging' as shown below:

If we wish to store information about the rent payment to be made by person (s) occupying different hotel rooms, then this information should appear as an attribute of

A. Person
B. Hotel Room
C. Lodging
D. None of these

gateit-2005 databases er-diagram easy

Answer key☟

3.10 Functional Dependency (3)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (10Q)

3.10.1 Functional Dependency: GATE CSE 2025 | Set 1 | Question: 37

Consider a relational schema team (name,city,owner), with functional dependencies $\{\text{name} \rightarrow \text{city}, \text{name} \rightarrow \text{owner}\}$.

The relation team is decomposed into two relations, $\text{t1}(\text{name,city})$ and $\text{t2}(\text{name, owner})$. Which of the following statement(s) is/are TRUE?

A. The relation team is NOT in BCNF .
B. The relations $t1$ and $t2$ are in BCNF.
C. The decomposition constitutes a lossless join.
D. The relation team is NOT in 3 NF .

gatecse2025-set1 databases functional-dependency database-normalization multiple-selects two-marks

Answer key☟

3.10.2 Functional Dependency: GATE CSE 2026 | Set 1 | Question: 20

Let P, Q, R and S be the attributes of a relation in a relational schema. Let $X \rightarrow Y$ indicate functional dependency in the context of a relational database, where $X, Y \subseteq \{P, Q, R, S\}$.

Which of the following options is/are always true?

A. If $(\{P, Q\} \rightarrow \{R\}$ and $\{P\} \rightarrow \{R\})$, then $\{Q\} \rightarrow \{R\}$
B. If $\{P, Q\} \rightarrow \{R\}$, then $(\{P\} \rightarrow \{R\}$ or $\{Q\} \rightarrow \{R\})$
C. If $(\{P\} \rightarrow \{R\}$ and $\{Q\} \rightarrow \{S\})$, then $\{P, Q\} \rightarrow \{R, S\}$
D. If $\{P\} \rightarrow \{R\}$, then $\{P, Q\} \rightarrow \{R\}$

gatecse-2026-set1 databases functional-dependency multiple-selects one-mark

Answer key

3.10.3 Functional Dependency: GATE DA 2025 | Question: 47

Consider a database relation R with attributes $\mathrm{ABCDEFG}$, and having the following functional dependencies:

$$\mathrm{A} \rightarrow \mathrm{BCEF} \quad \mathrm{E} \rightarrow \mathrm{DG} \quad \mathrm{BC} \rightarrow \mathrm{A}$$

Which of the following statements is/are correct?

A. A is the only candidate key of R
B. A, BC are the candidate keys of R
C. $\mathrm{A}, \mathrm{BC}, \mathrm{E}$ are the candidate keys of R
D. Relation R is not in Boyce-Codd Normal Form (BCNF)

gateda-2025 databases functional-dependency multiple-selects two-marks

Answer key

3.11 Indexing (15)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (7Q)

3.11.1 Indexing: GATE CSE 1989 | Question: 4-xiv

For secondary key processing which of the following file organizations is preferred? Give a one line justification:

A. Indexed sequential file organization.
B. Two-way linked list.
C. Inverted file organization.
D. Sequential file organization.

gate1989 normal databases indexing descriptive

Answer key

3.11.2 Indexing: GATE CSE 1990 | Question: 10b

One giga bytes of data are to be organized as an indexed-sequential file with a uniform blocking factor 8. Assuming a block size of 1 Kilo bytes and a block refrencing pointer size of 32 bits, find out the number of levels of indexing that would be required and the size of the index at each level. Determine also the size of the master index. The referencing capability (fanout ratio) per block of index storage may be considered to be 32.

gate1990 databases indexing descriptive

Answer key

3.11.3 Indexing: GATE CSE 1993 | Question: 14

An ISAM (indexed sequential) file consists of records of size 64 bytes each, including key field of size 14 bytes. An address of a disk block takes 2 bytes. If the disk block size is 512 bytes and there are $16K$ records, compute the size of the data and index areas in terms of number blocks. How many levels of tree do you have for the index?

gate1993 databases indexing normal descriptive

Answer key

3.11.4 Indexing: GATE CSE 1998 | Question: 1.35

There are five records in a database.

Name	Age	Occupation	Category
Rama	27	CON	A
Abdul	22	ENG	A
Jennifer	28	DOC	B
Maya	32	SER	D
Dev	24	MUS	C

There is an index file associated with this and it contains the values $1, 3, 2, 5$ and 4. Which one of the fields is the index built from?

A. Age
B. Name
C. Occupation
D. Category

gate1998 databases indexing normal

Answer key☟

3.11.5 Indexing: GATE CSE 2008 | Question: 16, ISRO2016-60

A clustering index is defined on the fields which are of type

A. non-key and ordering
B. non-key and non-ordering
C. key and ordering
D. key and non-ordering

gatecse-2008 easy databases indexing isro2016

Answer key☟

3.11.6 Indexing: GATE CSE 2008 | Question: 70

Consider a file of 16384 records. Each record is 32 bytes long and its key field is of size 6 bytes. The file is ordered on a non-key field, and the file organization is unspanned. The file is stored in a file system with block size 1024 bytes, and the size of a block pointer is 10 bytes. If the secondary index is built on the key field of the file, and a multi-level index scheme is used to store the secondary index, the number of first-level and second-level blocks in the multi-level index are respectively

A. 8 and 0
B. 128 and 6
C. 256 and 4
D. 512 and 5

gatecse-2008 databases indexing normal

Answer key☟

3.11.7 Indexing: GATE CSE 2011 | Question: 39

Consider a relational table r with sufficient number of records, having attributes $A_1, A_2, \dots, A_n$ and let $1 \leq p \leq n$. Two queries $Q1$ and $Q2$ are given below.

- $Q1 : \pi_{A_1,\dots,A_p}\left(\sigma_{A_p=c}\left(r\right)\right)$ where c is a constant
- $Q2 : \pi_{A_1,\dots,A_p}\left(\sigma_{c_1 \leq A_p \leq c_2}\left(r\right)\right)$ where c_1 and c_2 are constants.

The database can be configured to do ordered indexing on A_p or hashing on A_p. Which of the following statements is **TRUE**?

A. Ordered indexing will always outperform hashing for both queries
B. Hashing will always outperform ordered indexing for both queries
C. Hashing will outperform ordered indexing on $Q1$, but not on $Q2$
D. Hashing will outperform ordered indexing on $Q2$, but not on $Q1$

gatecse-2011 databases indexing normal

Answer key☟

3.11.8 Indexing: GATE CSE 2013 | Question: 15

An index is clustered, if

A. it is on a set of fields that form a candidate key

B. it is on a set of fields that include the primary key
C. the data records of the file are organized in the same order as the data entries of the index
D. the data records of the file are organized not in the same order as the data entries of the index

gatecse-2013 databases indexing normal

Answer key☟

3.11.9 Indexing: GATE CSE 2015 | Set 1 | Question: 24

A file is organized so that the ordering of the data records is the same as or close to the ordering of data entries in some index. Then that index is called

A. Dense B. Sparse C. Clustered D. Unclustered

gatecse-2015-set1 databases indexing easy

Answer key☟

3.11.10 Indexing: GATE CSE 2020 | Question: 54

Consider a database implemented using B^+ tree for file indexing and installed on a disk drive with block size of $4\,\text{KB}$. The size of search key is 12 bytes and the size of tree/disk pointer is 8 bytes. Assume that the database has one million records. Also assume that no node of the B^+ tree and no records are present initially in main memory. Consider that each record fits into one disk block. The minimum number of disk accesses required to retrieve any record in the database is _______

gatecse-2020 numerical-answers databases b-tree indexing two-marks

Answer key☟

3.11.11 Indexing: GATE CSE 2021 | Set 2 | Question: 21

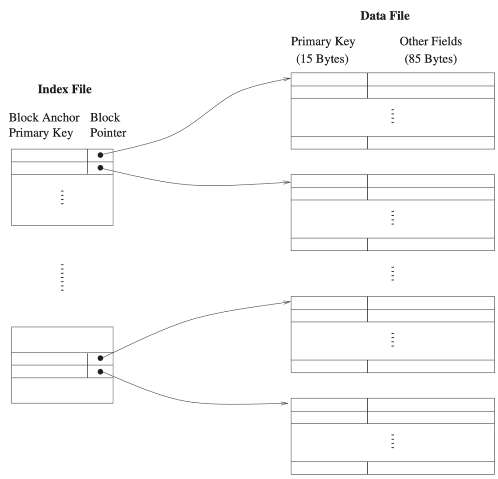
A data file consisting of $1,50,000$ student-records is stored on a hard disk with block size of 4096 bytes. The data file is sorted on the primary key RollNo. The size of a record pointer for this disk is 7 bytes. Each student-record has a candidate key attribute called ANum of size 12 bytes. Suppose an index file with records consisting of two fields, ANum value and the record pointer the corresponding student record, is built and stored on the same disk. Assume that the records of data file and index file are not split across disk blocks. The number of blocks in the index file is ________

gatecse-2021-set2 numerical-answers databases indexing one-mark

Answer key☟

3.11.12 Indexing: GATE CSE 2023 | Question: 52

Consider a database of fixed-length records, stored as an ordered file. The database has $25,000$ records, with each record being 100 bytes, of which the primary key occupies 15 bytes. The data file is block-aligned in that each data record is fully contained within a block. The database is indexed by a primary index file, which is also stored as a block-aligned ordered file. The figure below depicts this indexing scheme.



Suppose the block size of the file system is 1024 bytes, and a pointer to a block occupies 5 bytes. The system uses binary search on the index file to search for a record with a given key. You may assume that a binary search on an index file of b blocks takes $\lceil \log_2 b \rceil$ block accesses in the worst case.

Given a key, the number of block accesses required to identify the block in the data file that may contain a record with the key, in the worst case, is _____________.

gatecse-2023 databases file-system indexing numerical-answers two-marks

Answer key

3.11.13 Indexing: GATE CSE 2024 | Set 2 | Question: 16

Which of the following file organizations is/are $\mathrm{I/O}$ efficient for the scan operation in DBMS?

A. Sorted
B. Heap
C. Unclustered tree index
D. Unclustered hash index

gatecse-2024-set2 databases multiple-selects indexing one-mark

Answer key

3.11.14 Indexing: GATE CSE 2026 | Set 2 | Question: 36

An index in a DBMS is said to be dense if an index entry appears for every search-key value in the indexed file. Otherwise it is called a sparse index. Consider the following two statements.

$\mathrm{S1}$: A hash index must be a dense index
$\mathrm{S2}$: A B^+ tree index can be a sparse index

Which one of the following options is correct?

A. Both $\mathrm{S1}$ and $\mathrm{S2}$ are true
B. Both $\mathrm{S1}$ and $\mathrm{S2}$ are false
C. $\mathrm{S1}$ is true and $\mathrm{S2}$ is false
D. $\mathrm{S1}$ is false and $\mathrm{S2}$ is true

gatecse-2026-set2 databases indexing two-marks

Answer key

3.11.15 Indexing: GATE DA 2026 | Question: 22

In a relational database, a $\mathrm{B+}$ Tree Index is to be constructed for a relation on a key field. In a $\mathrm{B+}$ Tree, a Node Pointer points to a sub-tree and a Data Record Pointer points to a block of database records.

Let, Node size $= 4096$ bytes, Node Pointer size $= 10$ bytes, Search Key Field size $= 11$ bytes and Data Record Pointer size $= 12$ bytes.

The maximum number of Node Pointers that can be present in a non-leaf node of the $\mathrm{B+}$ Tree is _____. (Answer in integer)

gateda-2026 databases indexing b-tree numerical-answers one-mark

Answer key

3.12 Joins (7)

✍ **Practice Test:** Test 1 (7Q)

3.12.1 Joins: GATE CSE 2004 | Question: 14

Consider the following relation schema pertaining to a students database:

- Students (<u>rollno,</u> name, address)
- Enroll (<u>rollno, courseno</u>, coursename)

where the primary keys are shown underlined. The number of tuples in the student and Enroll tables are 120 and 8 respectively. What are the maximum and minimum number of tuples that can be present in (Student * Enroll), where '*' denotes natural join?

A. $8, 8$ B. $120, 8$ C. $960, 8$ D. $960, 120$

gatecse-2004 databases easy joins natural-join

Answer key

3.12.2 Joins: GATE CSE 2012 | Question: 50

Consider the following relations A, B and C :

A

ID	Name	Age
12	Arun	60
15	Shreya	24
99	Rohit	11

B

ID	Name	Age
15	Shreya	24
25	Hari	40
98	Rohit	20
99	Rohit	11

C

ID	Phone	Area
10	2200	02
99	2100	01

How many tuples does the result of the following relational algebra expression contain? Assume that the schema of $A \cup B$ is the same as that of A.

$$(A \cup B) \bowtie_{A.Id > 40 \vee C.Id < 15} C$$

A. 7 B. 4 C. 5 D. 9

gatecse-2012 databases joins normal

Answer key

3.12.3 Joins: GATE CSE 2014 | Set 2 | Question: 30

Consider a join (relation algebra) between relations $r(R)$ and $s(S)$ using the nested loop method. There are 3 buffers each of size equal to disk block size, out of which one buffer is reserved for intermediate results. Assuming $\text{size}(r(R)) < \text{size}(s(S))$, the join will have fewer number of disk block accesses if

A. relation $r(R)$ is in the outer loop.
B. relation $s(S)$ is in the outer loop.
C. join selection factor between $r(R)$ and $s(S)$ is more than 0.5.
D. join selection factor between $r(R)$ and $s(S)$ is less than 0.5.

gatecse-2014-set2 databases normal joins

Answer key

3.12.4 Joins: GATE IT 2005 | Question: 82a

A database table T_1 has 2000 records and occupies 80 disk blocks. Another table T_2 has 400 records and occupies 20 disk blocks. These two tables have to be joined as per a specified join condition that needs to be evaluated for every pair of records from these two tables. The memory buffer space available can hold exactly one block of records for T_1 and one block of records for T_2 simultaneously at any point in time. No index is available on either table.

If Nested-loop join algorithm is employed to perform the join, with the most appropriate choice of table to be used in outer loop, the number of block accesses required for reading the data are

A. 800000 B. 40080 C. 32020 D. 100

gateit-2005 databases normal joins

Answer key

3.12.5 Joins: GATE IT 2005 | Question: 82b

A database table T_1 has 2000 records and occupies 80 disk blocks. Another table T_2 has 400 records and occupies 20 disk blocks. These two tables have to be joined as per a specified join condition that needs to be evaluated for every pair of records from these two tables. The memory buffer space available can hold exactly one block of records for T_1 and one block of records for T_2 simultaneously at any point in time. No index is available on either table.

If, instead of Nested-loop join, Block nested-loop join is used, again with the most appropriate choice of table in the outer loop, the reduction in number of block accesses required for reading the data will be

A. 0 B. 30400 C. 38400 D. 798400

gateit-2005 databases normal joins

Answer key

3.12.6 Joins: GATE IT 2006 | Question: 14

Consider the relations $r_1(\text{P}, \text{Q}, \text{R})$ and $r_2(\text{R}, \text{S}, \text{T})$ with primary keys P and R respectively. The relation r_1 contains 2000 tuples and r_2 contains 2500 tuples. The maximum size of the join $r_1 \bowtie r_2$ is :

A. 2000 B. 2500 C. 4500 D. 5000

gateit-2006 databases joins natural-join normal

Answer key

3.12.7 Joins: GATE IT 2007 | Question: 68

Consider the following relation schemas :

- b-Schema = (b-name, b-city, assets)
- a-Schema = (a-num, b-name, bal)
- d-Schema = (c-name, a-number)

Let branch, account and depositor be respectively instances of the above schemas. Assume that account and depositor relations are much bigger than the branch relation.

Consider the following query:

$\Pi_{\text{c-name}}$ ($\sigma_{\text{b-city = "Agra"} \wedge \text{bal < 0}}$ (branch ⋈ (account ⋈ depositor)

Which one of the following queries is the most efficient version of the above query ?

A. $\Pi_{\text{c-name}}$ ($\sigma_{\text{bal < 0}}$ ($\sigma_{\text{b-city = "Agra"}}$ branch ⋈ account) ⋈ depositor)
B. $\Pi_{\text{c-name}}$ ($\sigma_{\text{b-city = "Agra"}}$ branch ⋈ ($\sigma_{\text{bal < 0}}$ account ⋈ depositor))
C. $\Pi_{\text{c-name}}$ (($\sigma_{\text{b-city = "Agra"}}$ branch ⋈ $\sigma_{\text{b-city = "Agra"} \wedge \text{bal < 0}}$ account) ⋈ depositor)
D. $\Pi_{\text{c-name}}$ ($\sigma_{\text{b-city = "Agra"}}$ branch ⋈ ($\sigma_{\text{b-city = "Agra"} \wedge \text{bal < 0}}$ account ⋈ depositor))

gateit-2007 databases joins relational-algebra normal

Answer key

3.13 Multivalued Dependency 4nf (1)

3.13.1 Multivalued Dependency 4nf: GATE IT 2007 | Question: 67

Consider the following implications relating to functional and multivalued dependencies given below, which may or may not be correct.

i. if $A \rightarrow\rightarrow B$ and $A \rightarrow\rightarrow C$ then $A \rightarrow BC$
ii. if $A \rightarrow B$ and $A \rightarrow C$ then $A \rightarrow\rightarrow BC$
iii. if $A \rightarrow\rightarrow BC$ and $A \rightarrow B$ then $A \rightarrow C$
iv. if $A \rightarrow BC$ and $A \rightarrow B$ then $A \rightarrow\rightarrow C$

Exactly how many of the above implications are valid?

A. 0 B. 1 C. 2 D. 3

gateit-2007 databases database-normalization multivalued-dependency-4nf normal

Answer key

3.14 Natural Join (3)

3.14.1 Natural Join: GATE CSE 2005 | Question: 30

Let r be a relation instance with schema R = (A, B, C, D). We define $r_1 = \pi_{A,B,C}(R)$ and $r_2 = \pi_{A,D}(r)$. Let $s = r_1 * r_2$ where $*$ denotes natural join. Given that the decomposition of r into r_1 and r_2 is lossy, which one of the following is TRUE?

A. $s \subset r$
B. $r \cup s = r$
C. $r \subset s$
D. $r * s = s$

gatecse-2005 databases relational-algebra natural-join normal

Answer key

3.14.2 Natural Join: GATE CSE 2010 | Question: 43

The following functional dependencies hold for relations $R(A, B, C)$ and $S(B, D, E)$.

- $B \rightarrow A$
- $A \rightarrow C$

The relation R contains 200 tuples and the relation S contains 100 tuples. What is the maximum number of tuples possible in the natural join $R \bowtie S$?

A. 100
B. 200
C. 300
D. 2000

gatecse-2010 databases normal natural-join

Answer key

3.14.3 Natural Join: GATE CSE 2015 | Set 2 | Question: 32

Consider two relations $R_1(A, B)$ with the tuples $(1,5), (3,7)$ and $R_2(A, C) = (1,7), (4,9)$.
Assume that $R(A, B, C)$ is the full natural outer join of R_1 and R_2. Consider the following tuples of the form (A, B, C) :
$a = (1, 5, null), b = (1, null, 7), c = (3, null, 9), d = (4, 7, null), e = (1, 5, 7),$
$f = (3, 7, null), g = (4, null, 9).$
Which one of the following statements is correct?

A. R contains a, b, e, f, g but not c, d.
B. R contains all a, b, c, d, e, f, g.
C. R contains e, f, g but not a, b.
D. R contains e but not f, g.

gatecse-2015-set2 databases normal natural-join

Answer key

3.15 Normal Forms (1)

✍ **Practice Test:** Test 1 (11Q)

3.15.1 Normal Forms: GATE DA 2026 | Question: 51

Consider an ER model with the entities $\mathrm{E1}$ (A_{11}, A_{12}, A_{13}) and $\mathrm{E2}$ (A_{21}, A_{22}, A_{23}), where, A_{11}, A_{12}, A_{13} are the attributes of $\mathrm{E1}$, and A_{21}, A_{22}, A_{23} are the attributes of $\mathrm{E2}$. Let A_{22} be a multi-valued attribute. A_{11} and A_{21} are the primary keys of $\mathrm{E1}$ and $\mathrm{E2}$, respectively.

Let $\mathrm{R12}$ be a many-to-many relationship between $\mathrm{E1}$ and $\mathrm{E2}$. Participation of both $\mathrm{E1}$ and $\mathrm{E2}$ in $\mathrm{R12}$ is total.

The minimum number of relations required to convert the ER model into relational model (assuming there is no other functional dependency) where each relation is in third normal form ($\mathrm{3NF}$) is _____. (Answer in **integer**)

gateda-2026 databases er-diagram normal-forms numerical-answers two-marks

Answer key

3.16 Query (3)

3.16.1 Query: GATE CSE 1997 | Question: 76-b

Consider the following relational database schema:

- EMP (eno name, age)
- PROJ (pno name)
- INVOLVED (eno, pno)

EMP contains information about employees. PROJ about projects and involved about which employees involved in which projects. The underlined attributes are the primary keys for the respective relations.

State in English (in not more than 15 words)

What the following relational algebra expressions are designed to determine

i. $\Pi_{eno}(\text{INVOLVED}) - \Pi_{eno}((\Pi_{eno}(\text{INVOLVED}) \times \Pi_{pno}(\text{PROJ})) - \text{INVOLVED})$
ii. $\Pi_{age}(\text{EMP}) - \Pi_{age}(\sigma_{E.age<Emp.age}((\rho E(\text{EMP}) \times \text{EMP}))$

(Note: $\rho E(\text{EMP})$ conceptually makes a copy of EMP and names it E (ρ is called the rename operator))

gate1997 databases sql descriptive normal relational-algebra query

Answer key☟

3.16.2 Query: GATE DA 2026 | Question: 41

Consider a table Employee(EmpID, TeamID), where the column EmpID (ID of an employee) is the primary key. The column TeamID denotes the team ID of the team of which the employee is a member. TeamID is a NOT NULL column.

We want to display the size of the team (denoted as TeamSize) in which each employee is a member by using SQL. As an example, the desired output for the given Employee table is also shown in tabular form.

Which of the following is/are correct?

Employee

EmpID	TeamID
1	8
2	8
3	8
4	7
5	7
6	9

Output

EmpID	TeamSize
1	3
2	3
3	3
4	2
5	2
6	1

A.
```
SELECT E.EmpID, B.TeamSize
FROM Employee AS E, (SELECT TeamID, COUNT(TeamID) AS
TeamSize FROM Employee GROUP BY TeamID) AS B
WHERE E.TeamID = B.TeamID
```

B.
```
SELECT A.EmpID, COUNT(B.TeamID) AS TeamSize
FROM Employee AS A, Employee AS B
WHERE A.TeamID = B.TeamID AND A.EmpID = B.EmpID
GROUP BY A.EmpID
```

C.
```
SELECT B.EmpID, B.TeamSize
FROM (SELECT EmpID, COUNT(TeamID) AS TeamSize
FROM Employee GROUP BY EmpID) AS B
```

D.
```
SELECT A.EmpID, B.TeamSize
FROM Employee AS A, (SELECT COUNT(TeamID) AS TeamSize FROM
Employee GROUP BY TeamID) AS B
WHERE A.TeamID = B.TeamID
```

gateda-2026 databases sql query multiple-selects two-marks

Answer key

3.16.3 Query: GATE DA 2026 | Question: 50

Let Account be a relation as shown.

Account

AccNo	Balance
A1	5000
A2	5000
A3	10000
A4	15000
A5	18000

Consider the given SQL query.

```
SELECT AccNo FROM Account AS A
WHERE (SELECT COUNT(*) FROM Account AS B
WHERE A.Balance < B.Balance) >= (SELECT COUNT(*)
FROM Account AS C WHERE A.Balance > C.Balance)
```

The number of rows returned by the SQL query is _____. (Answer in integer)

gateda-2026 databases sql query numerical-answers two-marks

Answer key

3.17 Referential Integrity (6)

✍ **Practice Test:** Test 1 (7Q)

3.17.1 Referential Integrity: GATE CSE 1997 | Question: 6.10, ISRO2016-54

Let $R(a, b, c)$ and $S(d, e, f)$ be two relations in which d is the foreign key of S that refers to the primary key of R. Consider the following four operations R and S

I. Insert into R
II. Insert into S
III. Delete from R
IV. Delete from S

Which of the following can cause violation of the referential integrity constraint above?

A. Both I and IV
B. Both II and III
C. All of these
D. None of these

gate1997 databases referential-integrity easy isro2016

Answer key

3.17.2 Referential Integrity: GATE CSE 2005 | Question: 76

The following table has two attributes A and C where A is the primary key and C is the foreign key referencing A with on-delete cascade.

A	C
2	4
3	4
4	3
5	2
7	2
9	5
6	4

The set of all tuples that must be additionally deleted to preserve referential integrity when the tuple $(2, 4)$ is deleted is:

A. $(3,4)$ and $(6,4)$
B. $(5,2)$ and $(7,2)$
C. $(5,2),(7,2)$ and $(9,5)$
D. $(3,4),(4,3)$ and $(6,4)$

gatecse-2005 databases referential-integrity normal

Answer key

3.17.3 Referential Integrity: GATE CSE 2012 | Question: 43

Suppose $R_1(\underline{A}, B)$ and $R_2(\underline{C}, D)$ are two relation schemas. Let r_1 and r_2 be the corresponding relation instances. B is a foreign key that refers to C in R_2. If data in r_1 and r_2 satisfy referential integrity constraints, which of the following is **ALWAYS TRUE**?

A. $\prod_B(r_1) - \prod_C(r_2) = \varnothing$
B. $\prod_C(r_2) - \prod_B(r_1) = \varnothing$
C. $\prod_B(r_1) = \prod_C(r_2)$
D. $\prod_B(r_1) - \prod_C(r_2) \neq \varnothing$

gatecse-2012 databases relational-algebra normal referential-integrity

Answer key

3.17.4 Referential Integrity: GATE CSE 2017 | Set 2 | Question: 19

Consider the following tables $T1$ and $T2$.

$T1$

P	Q
2	2
3	8
7	3
5	8
6	9
8	5
9	8

$T2$

R	S
2	2
8	3
3	2
9	7
5	7
7	2

In table $T1$ **P** is the primary key and **Q** is the foreign key referencing **R** in table $T2$ with on-delete cascade and on-update cascade. In table $T2$, **R** is the primary key and **S** is the foreign key referencing **P** in table $T1$ with on-delete set NULL and on-update cascade. In order to delete record $\langle 3, 8 \rangle$ from the table $T1$, the number of additional records that need to be deleted from table $T1$ is _______ gatecse-2017-set2 databases numerical-answers referential-integrity normal

Answer key

3.17.5 Referential Integrity: GATE CSE 2021 | Set 2 | Question: 6

Consider the following statements $S1$ and $S2$ about the relational data model:

- $S1$: A relation scheme can have at most one foreign key.
- $S2$: A foreign key in a relation scheme R cannot be used to refer to tuples of R.

Which one of the following choices is correct?

A. Both $S1$ and $S2$ are true
B. $S1$ is true and $S2$ is false
C. $S1$ is false and $S2$ is true
D. Both $S1$ and $S2$ are false

gatecse-2021-set2 databases referential-integrity one-mark

Answer key

3.17.6 Referential Integrity: GATE DA 2026 | Question: 16

Consider two relations r and s defined on the relational schemas $R(A, B)$ and $S(E, C)$, respectively. A is the primary key of R and E is a foreign key of S referencing A in R.

Which of the following operations will $\textbf{NEVER}$ violate the foreign key constraint?

A. Inserting records into relation r
B. Deleting records from relation s
C. Deleting records from relation r
D. Inserting records into relation s

gateda-2026 databases referential-integrity multiple-selects one-mark

Answer key

3.18 Relational Algebra (33)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (15Q) Test 4 (3Q)

3.18.1 Relational Algebra: GATE CSE 1992 | Question: 13b

Suppose we have a database consisting of the following three relations:

FREQUENTS	(CUSTOMER, HOTEL)
SERVES	(HOTEL, SNACKS)
LIKES	(CUSTOMER, SNACKS)

The first indicates the hotels each customer visits, the second tells which snacks each hotel serves and last indicates which snacks are liked by each customer. Express the following query in relational algebra:

Print the hotels the serve the snack that customer Rama likes.

gate1992 databases relational-algebra normal descriptive

Answer key

3.18.2 Relational Algebra: GATE CSE 1994 | Question: 13

Consider the following relational schema:

- COURSES (cno, cname)
- STUDENTS (rollno, sname, age, year)
- REGISTERED_FOR (cno, rollno)

The underlined attributes indicate the primary keys for the relations. The 'year' attribute for the STUDENTS relation indicates the year in which the student is currently studying (First year, Second year etc.)

A. Write a relational algebra query to print the roll number of students who have registered for cno 322.

B. Write a SQL query to print the age and year of the youngest student in each year.

gate1994 databases relational-algebra sql normal descriptive

Answer key

3.18.3 Relational Algebra: GATE CSE 1994 | Question: 3.8

Give a relational algebra expression using only the minimum number of operators from $(\cup, -)$ which is equivalent to $R \cap S$.

gate1994 databases relational-algebra normal descriptive

Answer key

3.18.4 Relational Algebra: GATE CSE 1995 | Question: 27

Consider the relation scheme.

AUTHOR	(ANAME, INSTITUTION, ACITY, AGE)
PUBLISHER	(PNAME, PCITY)
BOOK	(TITLE, ANAME, PNAME)

Express the following queries using (one or more of) SELECT, PROJECT, JOIN and DIVIDE operations.

A. Get the names of all publishers.
B. Get values of all attributes of all authors who have published a book for the publisher with PNAME='TECHNICAL PUBLISHERS'.
C. Get the names of all authors who have published a book for any publisher located in Madras

gate1995 databases relational-algebra normal descriptive

Answer key

3.18.5 Relational Algebra: GATE CSE 1996 | Question: 27

A library relational database system uses the following schema

- USERS (User#, User Name, Home Town)
- BOOKS (Book#, Book Title, Author Name)
- ISSUED (Book#, User#, Date)

Explain in one English sentence, what each of the following relational algebra queries is designed to determine

a. $\sigma_{\text{User\#}=6}\ (\pi_{\text{User\#, Book Title}}\ ((\text{USERS} \bowtie \text{ISSUED}) \bowtie \text{BOOKS}))$
b. $\pi_{\text{Author Name}}\ (\text{BOOKS} \bowtie \sigma_{\text{Home Town}=\text{Delhi}}\ (\text{USERS} \bowtie \text{ISSUED}))$

gate1996 databases relational-algebra descriptive

Answer key

3.18.6 Relational Algebra: GATE CSE 1997 | Question: 76-a

Consider the following relational database schema:

- EMP (eno name, age)
- PROJ (pno name)
- INVOLVED (eno, pno)

EMP contains information about employees. PROJ about projects and involved about which employees involved in which projects. The underlined attributes are the primary keys for the respective relations.

What is the relational algebra expression containing one or more of $\{\sigma, \pi, \times, \rho, -\}$ which is equivalent to SQL query.

```
select eno from EMP|INVOLVED where EMP.eno=INVOLVED.eno and INVOLVED.pno=3
```

gate1997 databases sql relational-algebra descriptive

Answer key

3.18.7 Relational Algebra: GATE CSE 1998 | Question: 1.33

Given two union compatible relations $R_1(A,B)$ and $R_2(C,D)$, what is the result of the operation $R_1 \bowtie_{A=C \wedge B=D} R_2$?

A. $R_1 \cup R_2$
B. $R_1 \times R_2$
C. $R_1 - R_2$
D. $R_1 \cap R_2$

gate1998 normal relational-algebra

Answer key

3.18.8 Relational Algebra: GATE CSE 1998 | Question: 27

Consider the following relational database schemes:

- COURSES (Cno, Name)
- PRE_REQ(Cno, Pre_Cno)
- COMPLETED (Student_no, Cno)

COURSES gives the number and name of all the available courses.

PRE_REQ gives the information about which courses are pre-requisites for a given course.

COMPLETED indicates what courses have been completed by students

Express the following using relational algebra:

List all the courses for which a student with Student_no 2310 has completed all the pre-requisites.

gate1998 databases relational-algebra normal descriptive

Answer key

3.18.9 Relational Algebra: GATE CSE 1999 | Question: 1.18, ISRO2016-53

Consider the join of a relation R with a relation S. If R has m tuples and S has n tuples then the maximum and minimum sizes of the join respectively are

A. $m+n$ and 0
B. mn and 0
C. $m+n$ and $|m-n|$
D. mn and $m+n$

gate1999 databases relational-algebra easy isro2016

Answer key

3.18.10 Relational Algebra: GATE CSE 2000 | Question: 1.23, ISRO2016-57

Given the relations

- employee (name, salary, dept-no), and
- department (dept-no, dept-name,address),

Which of the following queries cannot be expressed using the basic relational algebra operations $(\sigma, \pi, \times, \bowtie, \cup, \cap, -)$?

A. Department address of every employee
B. Employees whose name is the same as their department name
C. The sum of all employees' salaries
D. All employees of a given department

gatecse-2000 databases relational-algebra easy isro2016

Answer key

3.18.11 Relational Algebra: GATE CSE 2001 | Question: 1.24

Suppose the adjacency relation of vertices in a graph is represented in a table Adj (X, Y). Which of the following queries cannot be expressed by a relational algebra expression of constant length?

A. List all vertices adjacent to a given vertex
B. List all vertices which have self loops
C. List all vertices which belong to cycles of less than three vertices
D. List all vertices reachable from a given vertex

gatecse-2001 databases relational-algebra normal

Answer key

3.18.12 Relational Algebra: GATE CSE 2001 | Question: 1.25

Let r and s be two relations over the relation schemes R and S respectively, and let A be an attribute in R. The relational algebra expression $\sigma_{A=a}(r \bowtie s)$ is always equal to

A. $\sigma_{A=a}(r)$
B. r
C. $\sigma_{A=a}(r) \bowtie s$
D. None of the above

gatecse-2001 databases relational-algebra

Answer key

3.18.13 Relational Algebra: GATE CSE 2001 | Question: 21-a

Consider a relation examinee (regno, name, score), where regno is the primary key to score is a real number.

Write a relational algebra using $(\Pi, \sigma, \rho, \times)$ to find the list of names which appear more than once in examinee.

gatecse-2001 databases normal descriptive relational-algebra

Answer key

3.18.14 Relational Algebra: GATE CSE 2002 | Question: 15

A university placement center maintains a relational database of companies that interview students on campus and make job offers to those successful in the interview. The schema of the database is given below:

$\text{COMPANY}(\underline{\text{cname}}, \text{clocation})$	$\text{STUDENT}(\underline{\text{srollno}}, \text{sname}, \text{sdegree})$
$\text{INTERVIEW}(\underline{\text{cname}, \text{srollno}}, \text{idate})$	$\text{OFFER}(\underline{\text{cname}, \text{srollno}}, \text{osalary})$

The COMPANY relation gives the name and location of the company. The STUDENT relation gives the student's roll number, name and the degree program for which the student is registered in the university. The INTERVIEW relation gives the date on which a student is interviewed by a company. The OFFER relation gives the salary offered to a student who is successful in a company's interview. The key for each relation is indicated by the underlined attributes

a. Write a **relational algebra** expressions (using only the operators $\bowtie, \sigma, \pi, \cup, -$) for the following queries.
   i. List the $rollnumbers$ and $names$ of students who attended at least one interview but did not receive any job offer.
   ii. List the $rollnumbers$ and $names$ of students who went for interviews and received job offers from $every$ company with which they interviewed.
b. Write an SQL query to list, for each degree program in which more than $five$ students were offered jobs, the name of the degree and the average offered salary of students in this degree program.

gatecse-2002 databases normal descriptive relational-algebra sql

Answer key

3.18.15 Relational Algebra: GATE CSE 2003 | Question: 30

Consider the following SQL query

Select distinct $a_1, a_2, \ldots, a_n$

from $r_1, r_2, \ldots, r_m$

where P

For an arbitrary predicate P, this query is equivalent to which of the following relational algebra expressions?

A. $\Pi_{a_1,a_2,\ldots a_n} \sigma_p \left(r_1 \times r_2 \times \cdots \times r_m\right)$
B. $\Pi_{a_1,a_2,\ldots a_n} \sigma_p \left(r_1 \bowtie r_2 \bowtie \cdots \bowtie r_m\right)$
C. $\Pi_{a_1,a_2,\ldots a_n} \sigma_p \left(r_1 \cup r_2 \cup \cdots \cup r_m\right)$
D. $\Pi_{a_1,a_2,\ldots a_n} \sigma_p \left(r_1 \cap r_2 \cap \cdots \cap r_m\right)$

gatecse-2003 databases relational-algebra normal

Answer key

3.18.16 Relational Algebra: GATE CSE 2004 | Question: 51

Consider the relation Student (name, sex, marks), where the primary key is shown underlined, pertaining to students in a class that has at least one boy and one girl. What does the following relational algebra expression produce? (Note: ρ is the rename operator).

$\pi_{name}\{\sigma_{sex=female}(\text{Student})\} - \pi_{name}(\text{Student} \bowtie_{(sex=female \wedge x=male \wedge marks \leq m)} \rho_{n,x,m}(\text{Student}))$

A. names of girl students with the highest marks
B. names of girl students with more marks than some boy student
C. names of girl students with marks not less than some boy student
D. names of girl students with more marks than all the boy students

gatecse-2004 databases relational-algebra normal

Answer key

3.18.17 Relational Algebra: GATE CSE 2007 | Question: 59

Information about a collection of students is given by the relation $\text{studInfo}(\underline{\text{studId}}, \text{name}, \text{sex})$. The relation $\text{enroll}(\text{studId}, \text{courseId})$ gives which student has enrolled for (or taken) what course(s). Assume that every course is taken by at least one male and at least one female student. What does the following relational algebra expression represent?

$\pi_{courceId}\left(\left(\pi_{\text{studId}}\left(\sigma_{sex='female'}(\text{studInfo})\right) \times \pi_{courseId}(\text{enroll})\right) - \text{enroll}\right)$

A. Courses in which all the female students are enrolled.
B. Courses in which a proper subset of female students are enrolled.
C. Courses in which only male students are enrolled.
D. None of the above

gatecse-2007 databases relational-algebra normal

Answer key

3.18.18 Relational Algebra: GATE CSE 2008 | Question: 68

Let R and S be two relations with the following schema

$R(\underline{P, Q}, R1, R2, R3)$

$S(\underline{P, Q}, S1, S2)$

where $\{P, Q\}$ is the key for both schemas. Which of the following queries are equivalent?

I. $\Pi_P(R \bowtie S)$
II. $\Pi_P(R) \bowtie \Pi_P(S)$

III. $\Pi_P\left(\Pi_{P,Q}\left(R\right)\cap\Pi_{P,Q}\left(S\right)\right)$

IV. $\Pi_P\left(\Pi_{P,Q}\left(R\right)-\left(\Pi_{P,Q}\left(R\right)-\Pi_{P,Q}\left(S\right)\right)\right)$

A. Only I and II
B. Only I and III
C. Only I, II and III
D. Only I, III and IV

gatecse-2008 databases relational-algebra normal

Answer key

3.18.19 Relational Algebra: GATE CSE 2014 | Set 3 | Question: 21

What is the optimized version of the relation algebra expression $\pi_{A1}(\pi_{A2}(\sigma_{F1}(\sigma_{F2}(r))))$, where $A1, A2$ are sets of attributes in r with $A1 \subset A2$ and $F1, F2$ are Boolean expressions based on the attributes in r?

A. $\pi_{A1}(\sigma_{(F1\wedge F2)}(r))$
B. $\pi_{A1}(\sigma_{(F1\vee F2)}(r))$
C. $\pi_{A2}(\sigma_{(F1\wedge F2)}(r))$
D. $\pi_{A2}(\sigma_{(F1\vee F2)}(r))$

gatecse-2014-set3 databases relational-algebra easy

Answer key

3.18.20 Relational Algebra: GATE CSE 2014 | Set 3 | Question: 30

Consider the relational schema given below, where **eId** of the relation **dependent** is a foreign key referring to **empId** of the relation **employee**. Assume that every employee has at least one associated dependent in the **dependent** relation.

- **employee (empId, empName, empAge)**
- **dependent (depId, eId, depName, depAge)**

Consider the following relational algebra query:

$$\Pi_{empId}\left(employee\right)-\Pi_{empId}\left(employee \bowtie_{(empId=eID)\wedge(empAge\leq depAge)} dependent\right)$$

The above query evaluates to the set of **empId**s of employees whose age is greater than that of

A. some dependent.
B. all dependents.
C. some of his/her dependents.
D. all of his/her dependents.

gatecse-2014-set3 databases relational-algebra normal

Answer key

3.18.21 Relational Algebra: GATE CSE 2015 | Set 1 | Question: 7

SELECT operation in SQL is equivalent to

A. The selection operation in relational algebra
B. The selection operation in relational algebra, except that SELECT in SQL retains duplicates
C. The projection operation in relational algebra
D. The projection operation in relational algebra, except that SELECT in SQL retains duplicates

gatecse-2015-set1 databases sql relational-algebra easy

Answer key

3.18.22 Relational Algebra: GATE CSE 2017 | Set 1 | Question: 46

Consider a database that has the relation schema CR(StudentName, CourseName). An instance of the schema CR is as given below.

StudentName	CourseName
SA	CA
SA	CB
SA	CC
SB	CB
SB	CC
SC	CA
SC	CB
SC	CC
SD	CA
SD	CB
SD	CC
SD	CD
SE	CD
SE	CA
SE	CB
SF	CA
SF	CB
SF	CC

The following query is made on the database.

- $T1 \leftarrow \pi_{CourseName}\ (\sigma_{StudentName=SA}\ (CR))$
- $T2 \leftarrow CR \div T1$

The number of rows in $T2$ is _____________ .

gatecse-2017-set1 databases relational-algebra normal numerical-answers

Answer key

3.18.23 Relational Algebra: GATE CSE 2018 | Question: 41

Consider the relations $r(A, B)$ and $s(B, C)$, where $s.B$ is a primary key and $r.B$ is a foreign key referencing $s.B$. Consider the query

$Q : r \bowtie (\sigma_{B<5}(s))$

Let LOJ denote the natural left outer-join operation. Assume that r and s contain no null values.

Which of the following is NOT equivalent to Q?

A. $\sigma_{B<5}(r \bowtie s)$
B. $\sigma_{B<5}(r\ LOJ\ s)$
C. $r\ LOJ\ (\sigma_{B<5}(s))$
D. $\sigma_{B<5}(r)\ LOJ\ s$

gatecse-2018 databases relational-algebra normal two-marks

Answer key

3.18.24 Relational Algebra: GATE CSE 2019 | Question: 55

Consider the following relations $P(X, Y, Z), Q(X, Y, T)$ and $R(Y, V)$.

Table: P

X	Y	Z
X1	Y1	Z1
X1	Y1	Z2
X2	Y2	Z2
X2	Y4	Z4

Table: Q

X	Y	T
X2	Y1	2
X1	Y2	5
X1	Y1	6
X3	Y3	1

Table: R

Y	V
Y1	V1
Y3	V2
Y2	V3
Y2	V2

How many tuples will be returned by the following relational algebra query?

$\Pi_x(\sigma_{(P.Y=R.Y \wedge R.V=V2))}(P \times R)) - \Pi_x(\sigma_{(Q.Y=R.Y \wedge Q.T>2))}(Q \times R))$

Answer: ________

gatecse-2019 numerical-answers databases relational-algebra two-marks

Answer key

3.18.25 Relational Algebra: GATE CSE 2021 | Set 1 | Question: 27

The following relation records the age of 500 employees of a company, where $empNo$ (indicating the employee number) is the key:

$$empAge(\underline{empNo}, age)$$

Consider the following relational algebra expression:

$$\Pi_{empNo}(empAge \bowtie_{(age>age1)} \rho_{empNo1,age1}(empAge))$$

What does the above expression generate?

A. Employee numbers of only those employees whose age is the maximum
B. Employee numbers of only those employees whose age is more than the age of exactly one other employee
C. Employee numbers of all employees whose age is not the minimum
D. Employee numbers of all employees whose age is the minimum

gatecse-2021-set1 databases relational-algebra two-marks

Answer key

3.18.26 Relational Algebra: GATE CSE 2022 | Question: 15

Consider the following three relations in a relational database.

$\text{Employee}(\underline{\text{eId}}, \text{Name}),\ \text{Brand}(\underline{\text{bId}}, \text{bName}),\ \text{Own}(\underline{\text{eId}}, \underline{\text{bId}})$

Which of the following relational algebra expressions return the set of $eIds$ who own all the brands?

A. $\Pi_{eId}\,(\Pi_{eId,bId}(Own)/\Pi_{bId}(Brand))$
B. $\Pi_{eId}(Own) - \Pi_{eId}\,((\Pi_{eId}(Own) \times \Pi_{bId}(Brand)) - \Pi_{eId,bId}(Own))$
C. $\Pi_{eId}\,(\Pi_{eId,bId}(Own)/\Pi_{bId}(Own))$
D. $\Pi_{eId}\,((\Pi_{eId}(Own) \times \Pi_{bId}(Own))/\Pi_{bId}(Brand))$

gatecse-2022 databases relational-algebra multiple-selects one-mark

Answer key

3.18.27 Relational Algebra: GATE CSE 2024 | Set 1 | Question: 25

Consider the following two relations, $R(A, B)$ and $S(A, C)$:

R	
A	**B**
10	20
20	30
30	40
30	50
50	95

S	
A	**C**
10	90
30	45
40	80

The total number of tuples obtained by evaluating the following expression

$\sigma_{B<C}\left(R \bowtie_{R.A=S.A} S\right)$ is ___________.

gatecse-2024-set1 numerical-answers databases relational-algebra one-mark

Answer key

3.18.28 Relational Algebra: GATE CSE 2024 | Set 2 | Question: 35

The relation schema, Person (pid, city), describes the city of residence for every person uniquely identified by pid. The following relational algebra operators are available: selection, projection, cross product, and rename.

To find the list of cities where at least 3 persons reside, using the above operators, the minimum number of cross product operations that must be used is

A. 1 B. 2 C. 3 D. 4

gatecse-2024-set2 databases relational-algebra two-marks

Answer key

3.18.29 Relational Algebra: GATE DA 2025 | Question: 52

Consider the following tables, Loan and Borrower, of a bank.

Loan		
loan_number	**branch_name**	**amount**
L11	Banjara Hills	90000
L14	Kondapur	50000
L15	SR Nagar	40000
L22	SR Nagar	25000
L23	Balanagar	80000
L25	Kondapur	70000
L19	SR Nagar	65000

Borrower	
customer_name	**loan_num**
Anand	L11
Karteek	L11
Karteek	L14
Ankita	L15
Gopal	L19
Karteek	L22
Karteek	L23
Sunil	L23
Sunil	L25

Query: $\pi_{\text{branch\_name, customer\_name}}$ (**Loan** $\bowtie$ **Borrower**) $\div \pi_{\text{branch\_name}}$ (**Loan**) where $\bowtie$ denotes natural join.

The number of tuples returned by the above relational algebra query is ___________ (Answer in integer)

gateda-2025 databases relational-algebra numerical-answers two-marks

Answer key

3.18.30 Relational Algebra: GATE DA 2025 | Question: 7

Consider the following three relations:

Car (model, year, $\underline{\text{serial}}$, color)
Make (maker, $\underline{\text{model}}$)
Own ($\underline{\text{owner}}$, $\underline{\text{serial}}$)

A tuple in Car represents a specific car of a given model, made in a given year, with a serial number and a color. A tuple in Make specifies that a maker company makes cars of a certain model. A tuple in Own specifies that an owner owns the car with a given serial number. Keys are underlined; ($\underline{\text{owner}}$, $\underline{\text{serial}}$) together form key for Own. ($\bowtie$ denotes natural join)

$$\pi_{\text{owner}} \left(\text{Own} \bowtie \left(\sigma_{\text{color}=\text{"red"}} \left(\text{Car} \bowtie \left(\sigma_{\text{maker}=\text{"ABC"}} \text{ Make}\right)\right)\right)\right)$$

Which one of the following options describes what the above expression computes?

A. All owners of a red car, a car made by ABC, or a red car made by ABC
B. All owners of more than one car, where at least one car is red and made by ABC
C. All owners of a red car made by ABC
D. All red cars made by ABC

gateda-2025 databases relational-algebra one-mark

Answer key☟

3.18.31 Relational Algebra: GATE DA 2026 | Question: 32

Consider the given relations X, Y and Z. The relation X has three columns P, Q and R. The relation Y has three columns P, Q and S. The relation Z has two columns P and T.

X

P	Q	R
P1	Q1	R1
P2	Q2	R2
P3	Q3	R2

Y

P	Q	S
P1	Q1	2
P1	Q2	5
P2	Q1	6
P3	Q3	1

Z

P	T
P1	T1
P3	T2
P4	T3
P4	NULL

Consider the relational algebra expression

$$\pi_{P,R,S}\left[\left(\sigma_{(Q=Q3 \vee R=R2)}\left(X \bowtie Y\right)\right) \bowtie \left(\sigma_{(S>1)}\left(Y \bowtie Z\right)\right)\right]$$

where $\bowtie$ denotes natural join operation.

Which of the following options is the correct output for the given expression?

A. Two rows $(\text{P1}, \text{R1}, 2)$ and $(\text{P1}, \text{R1}, 5)$
B. Three rows $(\text{P1}, \text{R1}, 2), (\text{P1}, \text{R1}, 5)$ and $(\text{P2}, \text{R2}, 6)$
C. One row $(\text{P1}, \text{R1}, 2)$
D. Zero rows

gateda-2026 databases relational-algebra two-marks

Answer key

3.18.32 Relational Algebra: GATE DS&AI 2024 | Question: 16

Consider a database that includes the following relations:

Defender(name, rating, side, goals)
Forward(name, rating, assists, goals)
Team(name, club, price)

Which ONE of the following relational algebra expressions checks that every name occurring in Team appears in either Defender or Forward, where ϕ denotes the empty set?

A. Π_{name} (Team)$\setminus$ (Π_{name} (Defender)$\cap\Pi_{\text{name}}$ (Forward)) $=\phi$
B. (Π_{name} (Defender)$\cap\Pi_{\text{name}}$ (Forward)) $\setminus\Pi_{\text{name}}$ (Team)$=\phi$
C. Π_{name} (Team)$\setminus$ (Π_{name} (Defender)$\cup\Pi_{\text{name}}$ (Forward)) $=\phi$
D. (Π_{name} (Defender)$\cup\Pi_{\text{name}}$ (Forward)) $\setminus\Pi_{\text{name}}$ (Team)$=\phi$

gate-ds-ai-2024 relational-algebra databases one-mark

Answer key

3.18.33 Relational Algebra: GATE IT 2005 | Question: 68

A table 'student' with schema (roll, name, hostel, marks), and another table 'hobby' with schema (roll, hobbyname) contains records as shown below:

Table: student

Roll	Name	Hostel	Marks
1798	Manoj Rathor	7	95
2154	Soumic Banerjee	5	68
2369	Gumma Reddy	7	86
2581	Pradeep pendse	6	92
2643	Suhas Kulkarni	5	78
2711	Nitin Kadam	8	72
2872	Kiran Vora	5	92
2926	Manoj Kunkalikar	5	94
2959	Hemant Karkhanis	7	88
3125	Rajesh Doshi	5	82

Table: hobby

Roll	Hobby Name
1798	chess
1798	music
2154	music
2369	swimming
2581	cricket
2643	chess
2643	hockey
2711	volleyball
2872	football
2926	cricket
2959	photography
3125	music
3125	chess

The following SQL query is executed on the above tables:

```
select hostel
from student natural join hobby
where marks >= 75 and roll between 2000 and 3000;
```

Relations S and H with the same schema as those of these two tables respectively contain the same information as tuples. A new relation S' is obtained by the following relational algebra operation:

$$S' = \Pi_{\text{hostel}}\left(\left(\sigma_{s.roll=H.roll}\left(\sigma_{marks>75 \text{ and } roll>2000 \text{ and } roll<3000}\quad(S)\right) \times (H)\right)\right.$$

The difference between the number of rows output by the SQL statement and the number of tuples in S' is

A. 6 B. 4 C. 2 D. 0

gateit-2005 databases sql relational-algebra normal

Answer key

3.19 Relational Calculus (13)

✍ **Practice Test:** Test 1 (13Q)

3.19.1 Relational Calculus: GATE CSE 1993 | Question: 23

The following relations are used to store data about students, courses, enrollment of students in courses and teachers of courses. Attributes for primary key in each relation are marked by '*'.

```
Students (rollno*, sname, saddr)
courses (cno*, cname)
enroll(rollno*, cno*, grade)
teach(tno*, tname, cao*)
```

(cno is course number cname is course name, tno is teacher number, tname is teacher name, sname is student name, etc.)

Write a SQL query for retrieving roll number and name of students who got A grade in at least one course taught by teacher names Ramesh for the above relational database.

gate1993 databases sql relational-calculus normal descriptive

Answer key

3.19.2 Relational Calculus: GATE CSE 1998 | Question: 2.19

Which of the following query transformations (i.e., replacing the l.h.s. expression by the r.h.s expression) is incorrect? R1 and R2 are relations, C1 and C2 are selection conditions and A1 and A2 are attributes of R1.

A. $\sigma_{C_1}\left(\sigma_{C_2}\left(R_1\right)\right) \rightarrow \sigma_{C_2}\left(\sigma_{C_1}\left(R_1\right)\right)$
B. $\sigma_{C_1}\left(\pi_{A_1}\left(R_1\right)\right) \rightarrow \pi_{A_1}\left(\sigma_{C_1}\left(R_1\right)\right)$
C. $\sigma_{C_1}\left(R_1 \cup R_2\right) \rightarrow \sigma_{C_1}\left(R_1\right) \cup \sigma_{C_1}\left(R_2\right)$
D. $\pi_{A_1}\left(\sigma_{C_1}\left(R_1\right)\right) \rightarrow \sigma_{C_1}\left(\pi_{A_1}\left(R_1\right)\right)$

gate1998 databases relational-calculus normal

Answer key

3.19.3 Relational Calculus: GATE CSE 1999 | Question: 1.19

The relational algebra expression equivalent to the following tuple calculus expression:

$\{t \mid t \in r \wedge (t[A]=10 \wedge t[B]=20)\}$ is

A. $\sigma_{(A=10 \vee B=20)}(r)$
B. $\sigma_{(A=10)}(r) \cup \sigma_{(B=20)}(r)$
C. $\sigma_{(A=10)}(r) \cap \sigma_{(B=20)}(r)$
D. $\sigma_{(A=10)}(r) - \sigma_{(B=20)}(r)$

gate1999 databases relational-calculus normal

Answer key

3.19.4 Relational Calculus: GATE CSE 2001 | Question: 2.24

Which of the following relational calculus expression is not safe?

A. $\{t \mid \exists u \in R_1\,(t[A]=u[A]) \wedge \neg \exists s \in R_2\,(t[A]=s[A])\}$
B. $\{t \mid \forall u \in R_1\,(u[A]=\text{"}x\text{"} \Rightarrow \exists s \in R_2\,(t[A]=s[A] \wedge s[A]=u[A]))\}$
C. $\{t \mid \neg(t \in R_1)\}$
D. $\{t \mid \exists u \in R_1\,(t[A]=u[A]) \wedge \exists s \in R_2\,(t[A]=s[A])\}$

gatecse-2001 relational-calculus normal databases

Answer key

3.19.5 Relational Calculus: GATE CSE 2002 | Question: 1.20

With regards to the expressive power of the formal relational query languages, which of the following statements is true?

A. Relational algebra is more powerful than relational calculus
B. Relational algebra has the same power as relational calculus
C. Relational algebra has the same power as safe relational calculus
D. None of the above

gatecse-2002 databases relational-calculus normal

Answer key

3.19.6 Relational Calculus: GATE CSE 2004 | Question: 13

Let $R_1\left(\underline{A}, B, C\right)$ and $R_2\left(\underline{D}, E\right)$ be two relation schema, where the primary keys are shown underlined, and let C be a foreign key in R_1 referring to R_2. Suppose there is no violation of the above referential integrity constraint in the corresponding relation instances r_1 and r_2. Which of the following relational algebra expressions would necessarily produce an empty relation?

A. $\Pi_D(r_2) - \Pi_C(r_1)$
B. $\Pi_C(r_1) - \Pi_D(r_2)$
C. $\Pi_D\left(r_1 \bowtie_{C \neq D} r_2\right)$
D. $\Pi_C\left(r_1 \bowtie_{C=D} r_2\right)$

gatecse-2004 databases relational-calculus easy

Answer key

3.19.7 Relational Calculus: GATE CSE 2007 | Question: 60

Consider the relation **employee**(name, sex, supervisorName) with *name* as the key, *supervisorName* gives the name of the supervisor of the employee under consideration. What does the following Tuple Relational Calculus query produce?

$$\{e.name \mid employee(e) \wedge (\forall x)\left[\neg employee\left(x\right) \vee x.supervisorName \neq e.name \vee x.sex = \text{``}male\text{''}\right]\}$$

A. Names of employees with a male supervisor.
B. Names of employees with no immediate male subordinates.
C. Names of employees with no immediate female subordinates.
D. Names of employees with a female supervisor.

gatecse-2007 databases relational-calculus normal

Answer key

3.19.8 Relational Calculus: GATE CSE 2008 | Question: 15

Which of the following tuple relational calculus expression(s) is/are equivalent to $\forall t \in r\left(P\left(t\right)\right)$?

I. $\neg \exists t \in r\left(P\left(t\right)\right)$
II. $\exists t \notin r\left(P\left(t\right)\right)$
III. $\neg \exists t \in r\left(\neg P\left(t\right)\right)$
IV. $\exists t \notin r\left(\neg P\left(t\right)\right)$

A. I only
B. II only
C. III only
D. III and IV only

gatecse-2008 databases relational-calculus normal

Answer key

3.19.9 Relational Calculus: GATE CSE 2009 | Question: 45

Let R and S be relational schemes such that $R = \{a, b, c\}$ and $S = \{c\}$. Now consider the following queries on the database:

1. $\pi_{R-S}(r) - \pi_{R-S}\left(\pi_{R-S}(r) \times s - \pi_{R-S,S}(r)\right)$
2. $\{t \mid t \in \pi_{R-S}(r) \wedge \forall u \in s\left(\exists v \in r\left(u = v[S] \wedge t = v\left[R - S\right]\right)\right)\}$

3. $\{t \mid t \in \pi_{R-S}(r) \wedge \forall v \in r\,(\exists u \in s\,(u = v[S] \wedge t = v\,[R - S]))\}$

4.
```
Select R.a,R.b
  From R,S
  Where R.c = S.c
```

Which of the above queries are equivalent?

A. 1 and 2 B. 1 and 3 C. 2 and 4 D. 3 and 4

gatecse-2009 databases relational-calculus difficult

Answer key

3.19.10 Relational Calculus: GATE CSE 2013 | Question: 35

Consider the following relational schema.

- Students(rollno: integer, sname: string)
- Courses(courseno: integer, cname: string)
- Registration(rollno: integer, courseno: integer, percent: real)

Which of the following queries are equivalent to this query in English?

"Find the distinct names of all students who score more than 90% *in the course numbered 107"*

I.
```
SELECT DISTINCT S.sname FROM Students as S, Registration
  as R WHERE
  R.rollno=S.rollno AND R.courseno=107 AND R.percent >90
```

II. $\prod_{sname}(\sigma_{courseno=107 \wedge percent>90}\,(Registration \bowtie Students))$

III. $\{T \mid \exists S \in Students, \exists R \in Registration(S.rollno = R.rollno$
$\wedge R.courseno = 107 \wedge R.percent > 90 \wedge T.sname = S.sname)\}$

IV.
$$\{\langle S_N \rangle \mid \exists S_R \exists R_P (\langle S_R, S_N \rangle \in Students \wedge$$
$$\langle S_R, 107, R_P \rangle \in Registration \wedge R_P > 90)\}$$

A. I, II, III and IV
B. I, II and III only
C. I, II and IV only
D. II, III and IV only

gatecse-2013 databases sql relational-calculus normal

Answer key

3.19.11 Relational Calculus: GATE IT 2006 | Question: 15

Which of the following relational query languages have the same expressive power?

I. Relational algebra
II. Tuple relational calculus restricted to safe expressions
III. Domain relational calculus restricted to safe expressions

A. II and III only B. I and II only C. I and III only D. I, II and III

gateit-2006 databases relational-algebra relational-calculus easy

Answer key

3.19.12 Relational Calculus: GATE IT 2007 | Question: 65

Consider a selection of the form $\sigma_{A \leq 100}(r)$, where r is a relation with 1000 tuples. Assume that the attribute values for A among the tuples are uniformly distributed in the interval $[0, 500]$. Which one of the following options is the best estimate of the number of tuples returned by the given selection query ?

A. 50 B. 100 C. 150 D. 200

gateit-2007 databases relational-calculus probability normal

Answer key

3.19.13 Relational Calculus: GATE IT 2008 | Question: 75

Consider the following relational schema:

- $\text{Student}(\underline{\text{school-id}, \text{sch-roll-no}}, \text{sname}, \text{saddress})$
- $\text{School}(\underline{\text{school-id}}, \text{sch-name}, \text{sch-address}, \text{sch-phone})$
- $\text{Enrolment}(\underline{\text{school-id}, \text{sch-roll-no}}, \text{erollno}, \text{examname})$
- $\text{ExamResult}(\underline{\text{erollno}, \text{examname}}, \text{marks})$

Consider the following tuple relational calculus query.

$$\{t \mid \exists E \in \text{Enrolment} \quad t = E.\text{school-id} \wedge |\{x \mid x \in \text{Enrolment} \wedge x.\text{school-id} = t \wedge (\exists B \in \text{ExamResult} \quad B.\text{er}$$

If a student needs to score more than 35 marks to pass an exam, what does the query return?

A. The empty set
B. schools with more than 35% of its students enrolled in some exam or the other
C. schools with a pass percentage above 35% over all exams taken together
D. schools with a pass percentage above 35% over each exam

gateit-2008 databases relational-calculus normal

Answer key

3.20 Relational Model (2)

3.20.1 Relational Model: GATE CSE 2023 | Question: 6

Which one of the options given below refers to the degree (or arity) of a relation in relational database systems?

A. Number of attributes of its relation schema.
B. Number of tuples stored in the relation.
C. Number of entries in the relation.
D. Number of distinct domains of its relation schema.

gatecse-2023 databases relational-model one-mark easy

Answer key

3.20.2 Relational Model: GATE DA 2025 | Question: 46

Consider the following two relations, named Customer and Person, in a database:

```
Person (
aadhaar CHAR(12) PRIMARY KEY,
name VARCHAR(32));

Customer (
name VARCHAR (32),
email VARCHAR(32) PRIMARY KEY,
phone CHAR(10),
aadhaar CHAR(12),
  FOREIGN KEY (aadhaar) REFERENCES Person(aadhaar));
```

Which of the following statements is/are correct?

A. aadhaar is a candidate key in the Customer relation
B. phone can be NULL in the Customer relation
C. aadhaar is a candidate key in the Person relation
D. aadhaar can be NULL in the Person relation

gateda-2025 databases relational-model multiple-selects two-marks

Answer key

3.21 SQL (58)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (15Q) Test 4 (15Q) Test 5 (15Q) Test 6 (4Q)

3.21.1 SQL: GATE CSE 1988 | Question: 12iii

Describe the relational algebraic expression giving the relation returned by the following SQL query.

```
Select    SNAME
from      S
Where     SNOin
          (select   SNO
          from      SP
          where     PNOin
                    (select   PNO
                    from      P
                    Where     COLOUR='BLUE'))
```

gate1988 normal descriptive databases sql

Answer key

3.21.2 SQL: GATE CSE 1988 | Question: 12iv

```
Select    SNAME
from      S
Where     SNOin
          (select   SNO
          from      SP
          where     PNOin
                    (select   PNO
                    from      P
                    Where     COLOUR='BLUE'))
```

What relations are being used in the above SQL query? Given at least two attributes of each of these relations.

gate1988 normal descriptive databases sql

Answer key

3.21.3 SQL: GATE CSE 1990 | Question: 10-a

Consider the following relational database:

- employees (eno, ename, address, basic-salary)
- projects (pno, pname, nos-of-staffs-allotted)
- working (pno, eno, pjob)

The queries regarding data in the above database are formulated below in SQL. Describe in ENGLISH sentences the two queries that have been posted:

i.
```
SELECT ename
FROM employees
WHERE eno IN
  (SELECT eno
  FROM working
  GROUP BY eno
  HAVING COUNT(*)=
    (SELECT COUNT(*)
    FROM projects))
```

ii.
```
SELECT pname
FROM projects
WHERE pno IN
  (SELECT pno
  FROM projects
  MINUS
  SELECT DISTINCT pno
  FROM working);
```

gate1990 descriptive databases sql

Answer key

3.21.4 SQL: GATE CSE 1991 | Question: 12,b

Suppose a database consist of the following relations:

```
SUPPLIER (SCODE,SNAME,CITY).

PART (PCODE,PNAME,PDESC,CITY).

PROJECTS (PRCODE,PRNAME,PRCITY).

SPPR (SCODE,PCODE,PRCODE,QTY).
```

Write algebraic solution to the following :

i. Get SCODE values for suppliers who supply to both projects PR1 and PR2.
ii. Get PRCODE values for projects supplied by at least one supplier not in the same city.

sql gate1991 normal databases descriptive

Answer key

3.21.5 SQL: GATE CSE 1991 | Question: 12-a

Suppose a database consist of the following relations:

```
SUPPLIER (SCODE,SNAME,CITY).
PART (PCODE,PNAME,PDESC,CITY).
PROJECTS (PRCODE,PRNAME,PRCITY).
SPPR (SCODE,PCODE,PRCODE,QTY).
```

Write SQL programs corresponding to the following queries:

i. Print PCODE values for parts supplied to any project in DEHLI by a supplier in DELHI.
ii. Print all triples <CITY, PCODE, CITY> such that a supplier in first city supplies the specified part to a project in the second city, but do not print the triples in which the two CITY values are same.

gate1991 databases sql normal descriptive

Answer key

3.21.6 SQL: GATE CSE 1993 | Question: 24

The following relations are used to store data about students, courses, enrollment of students in courses and teachers of courses. Attributes for primary key in each relation are marked by '*'.

- $\text{students}(\text{rollno}^*, \text{sname}, \text{saddr})$
- $\text{courses}(\text{cno}^*, \text{cname})$
- $\text{enroll}(\text{rollno}^*, \text{cno}^*, \text{grade})$
- $\text{teach}(\text{tno}^*, \text{tname}, \text{cao}^*)$

(cno is course number, cname is course name, tno is teacher number, tname is teacher name, sname is student name, etc.)

For the relational database given above, the following functional dependencies hold:

- $\text{rollno} \rightarrow \text{sname}, \text{saddr}$
- $\text{cno} \rightarrow \text{cname}$
- $\text{tno} \rightarrow \text{tname}$
- $\text{rollno}, \text{cno} \rightarrow \text{grade}$

a. Is the database in 3^{rd} normal form $(3NF)$?
b. If yes, prove that it is in 3NF. If not, normalize the relations so that they are in 3NF (without proving).

gate1993 databases sql normal descriptive database-normalization

Answer key

3.21.7 SQL: GATE CSE 1998 | Question: 7-a

Suppose we have a database consisting of the following three relations.

- $\text{FREQUENTS (student, parlor)}$ giving the parlors each student visits.
- $\text{SERVES (parlor, ice-cream)}$ indicating what kind of ice-creams each parlor serves.
- $\text{LIKES (student, ice-cream)}$ indicating what ice-creams each student likes.

(Assume that each student likes at least one ice-cream and frequents at least one parlor)

Express the following in SQL:

Print the students that frequent at least one parlor that serves some ice-cream that they like.

gate1998 databases sql descriptive

Answer key

3.21.8 SQL: GATE CSE 1999 | Question: 2.25

Which of the following is/are correct?

A. An SQL query automatically eliminates duplicates
B. An SQL query will not work if there are no indexes on the relations
C. SQL permits attribute names to be repeated in the same relation
D. None of the above

gate1999 databases sql easy

Answer key

3.21.9 SQL: GATE CSE 1999 | Question: 22-a

Consider the set of relations

- EMP (Employee-no. Dept-no, Employee-name, Salary)
- DEPT (Dept-no. Dept-name, Location)

Write an SQL query to:

a. Find all employees names who work in departments located at 'Calcutta' and whose salary is greater than Rs.50,000.
b. Calculate, for each department number, the number of employees with a salary greater than Rs. 1,00,000.

gate1999 databases sql easy descriptive

Answer key

3.21.10 SQL: GATE CSE 1999 | Question: 22-b

Consider the set of relations

- EMP (Employee-no. Dept-no, Employee-name, Salary)
- DEPT (Dept-no. Dept-name, Location)

Write an SQL query to:

Calculate, for each department number, the number of employees with a salary greater than Rs. 1,00,000

gate1999 databases sql descriptive easy

Answer key

3.21.11 SQL: GATE CSE 2000 | Question: 2.25

Given relations r(w, x) and s(y, z) the result of

```
select distinct w, x
from r, s
```

is guaranteed to be same as r, provided.

A. r has no duplicates and s is non-empty
B. r and s have no duplicates
C. s has no duplicates and r is non-empty
D. r and s have the same number of tuples

gatecse-2000 databases sql

Answer key

3.21.12 SQL: GATE CSE 2000 | Question: 2.26

In SQL, relations can contain null values, and comparisons with null values are treated as unknown. Suppose all comparisons with a null value are treated as false. Which of the following pairs is not equivalent?

A. $x = 5 \quad not(not(x = 5))$
B. $x = 5 \quad x > 4$ and $x < 6,$ where x is an integer
C. $x \neq 5 \quad not(x = 5)$
D. none of the above

gatecse-2000 databases sql normal

Answer key

3.21.13 SQL: GATE CSE 2000 | Question: 22

Consider a bank database with only one relation

transaction (transno, acctno, date, amount)

The amount attribute value is positive for deposits and negative for withdrawals.

a. Define an SQL view TP containing the information
(acctno,T1.date,T2.amount)
for every pair of transaction T1,T2 and such that T1 and T2 are transaction on the same account and the date of T2 is $\leq$ the date of T1.
b. Using only the above view TP, write a query to find for each account the minimum balance it ever reached (not including the 0 balance when the account is created). Assume there is at most one transaction per day on each account and each account has at least one transaction since it was created. To simplify your query, break it up into 2 steps by defining an intermediate view V.

gatecse-2000 databases sql normal descriptive

Answer key

3.21.14 SQL: GATE CSE 2001 | Question: 2.25

Consider a relation geq which represents "greater than or equal to", that is, $(x, y) \in$ geq only if $y \geq x$.

```
create table geq
(
   ib integer not null,
   ub integer not null,
   primary key ib,
   foreign key (ub) references geq on delete cascade
);
```

Which of the following is possible if tuple (x,y) is deleted?

A. A tuple (z,w) with z > y is deleted
B. A tuple (z,w) with z > x is deleted
C. A tuple (z,w) with w < x is deleted
D. The deletion of (x,y) is prohibited

gatecse-2001 databases sql normal

Answer key

3.21.15 SQL: GATE CSE 2001 | Question: 21-b

Consider a relation examinee (regno, name, score), where regno is the primary key to score is a real number.

Write an SQL query to list the *regno* of examinees who have a score greater than the average score.

gatecse-2001 databases sql normal descriptive

Answer key

3.21.16 SQL: GATE CSE 2001 | Question: 21-c

Consider a relation $\text{examinee (regno, name, score)}$, where regno is the primary key to score is a real number.

Suppose the relation $\text{appears (regno, centr\_code)}$ specifies the center where an examinee appears. Write an SQL query to list the centr_code having an examinee of score greater than 80.

gatecse-2001 databases sql normal descriptive

Answer key

3.21.17 SQL: GATE CSE 2003 | Question: 86

Consider the set of relations shown below and the SQL query that follows.

Students: (Roll_number, Name, Date_of_birth)

Courses: (Course_number, Course_name, Instructor)

Grades: (Roll_number, Course_number, Grade)

```
Select distinct Name
from Students, Courses, Grades
where Students.Roll_number=Grades.Roll_number
 and Courses.Instructor = 'Korth'
 and Courses.Course_number = Grades.Course_number
 and Grades.Grade = 'A'
```

Which of the following sets is computed by the above query?

A. Names of students who have got an A grade in all courses taught by Korth
B. Names of students who have got an A grade in all courses
C. Names of students who have got an A grade in at least one of the courses taught by Korth
D. None of the above

gatecse-2003 databases sql easy

Answer key

3.21.18 SQL: GATE CSE 2004 | Question: 53

The employee information in a company is stored in the relation

- Employee (name, sex, salary, deptName)

Consider the following SQL query

```
Select deptName
   From Employee
   Where sex = 'M'
   Group by deptName
   Having avg(salary) >
      (select avg (salary) from Employee)
```

It returns the names of the department in which

A. the average salary is more than the average salary in the company
B. the average salary of male employees is more than the average salary of all male employees in the company
C. the average salary of male employees is more than the average salary of employees in same the department
D. the average salary of male employees is more than the average salary in the company

gatecse-2004 databases sql normal

Answer key

3.21.19 SQL: GATE CSE 2005 | Question: 77, ISRO2016-55

The relation **book** (title, price) contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL query list?

```
select title
from book as B
where (select count(*)
  from book as T
  where T.price>B.price) < 5
```

A. Titles of the four most expensive books
B. Title of the fifth most inexpensive book
C. Title of the fifth most expensive book
D. Titles of the five most expensive books

gatecse-2005 databases sql easy isro2016

Answer key

3.21.20 SQL: GATE CSE 2006 | Question: 67

Consider the relation account (customer, balance) where the customer is a primary key and there are no null values. We would like to rank customers according to decreasing balance. The customer with the largest balance gets rank 1. Ties are not broke but ranks are skipped: if exactly two customers have the largest balance they each get rank 1 and rank 2 is not assigned.

Query1:

```
select A.customer, count(B.customer)
from account A, account B
where A.balance <=B.balance
group by A.customer
```

Query2:

```
select A.customer, 1+count(B.customer)
from account A, account B
where A.balance < B.balance
group by A.customer
```

Consider these statements about Query1 and Query2.

1. Query1 will produce the same row set as Query2 for some but not all databases.
2. Both Query1 and Query 2 are a correct implementation of the specification
3. Query1 is a correct implementation of the specification but Query2 is not
4. Neither Query1 nor Query2 is a correct implementation of the specification
5. Assigning rank with a pure relational query takes less time than scanning in decreasing balance order assigning ranks using ODBC.

Which two of the above statements are correct?

A. 2 and 5 B. 1 and 3 C. 1 and 4 D. 3 and 5

gatecse-2006 databases sql normal

Answer key

3.21.21 SQL: GATE CSE 2006 | Question: 68

Consider the relation enrolled (student, course) in which (student, course) is the primary key, and the relation paid (student, amount) where student is the primary key. Assume no null values and no foreign keys or integrity constraints.
Given the following four queries:

Query1:

```
select student from enrolled where student in (select student from paid)
```

Query2:

```
select student from paid where student in (select student from enrolled)
```

Query3:

```
select E.student from enrolled E, paid P where E.student = P.student
```

Query4:

```
select student from paid where exists
  (select * from enrolled where enrolled.student = paid.student)
```

Which one of the following statements is correct?

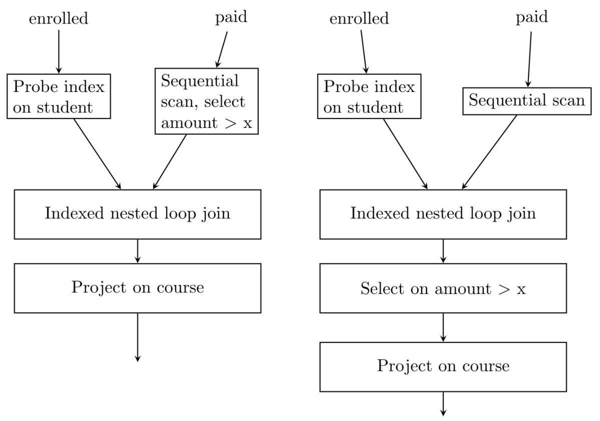
A. All queries return identical row sets for any database
B. Query2 and Query4 return identical row sets for all databases but there exist databases for which Query1 and Query2 return different row sets
C. There exist databases for which Query3 returns strictly fewer rows than Query2
D. There exist databases for which Query4 will encounter an integrity violation at runtime

gatecse-2006 databases sql normal

Answer key☟

3.21.22 SQL: GATE CSE 2006 | Question: 69

Consider the relation enrolled (student, course) in which (student, course) is the primary key, and the relation paid (student, amount) where student is the primary key. Assume no null values and no foreign keys or integrity constraints. Assume that amounts $6000, 7000, 8000, 9000$ and 10000 were each paid by 20% of the students. Consider these query plans (Plan 1 on left, Plan 2 on right) to *"list all courses taken by students who have paid more than x"*



A disk seek takes $4ms$, disk data transfer bandwidth is 300 MB/s and checking a tuple to see if amount is greater than x takes $10\mu s$. Which of the following statements is correct?

A. Plan 1 and Plan 2 will not output identical row sets for all databases
B. A course may be listed more than once in the output of Plan 1 for some databases
C. For $x = 5000$, Plan 1 executes faster than Plan 2 for all databases
D. For $x = 9000$, Plan I executes slower than Plan 2 for all databases

gatecse-2006 databases sql normal

Answer key☟

3.21.23 SQL: GATE CSE 2007 | Question: 61

Consider the table **employee**(empId, name, department, salary) and the two queries Q_1, Q_2 below. Assuming that department 5 has more than one employee, and we want to find the employees who get higher salary than anyone in the department 5, which one of the statements is **TRUE** for any arbitrary employee table?

Q_1 :
```
Select e.empId
From employee e
Where not exists
   (Select * From employee s Where s.department = "5" and s.salary >= e.salary)
```

Q_2 :
```
Select e.empId
From employee e
Where e.salary > Any
   (Select distinct salary From employee s Where s.department = "5")
```

A. Q_1 is the correct query
B. Q_2 is the correct query
C. Both Q_1 and Q_2 produce the same answer
D. Neither Q_1 nor Q_2 is the correct query

gatecse-2007 databases sql normal verbal-aptitude

Answer key

3.21.24 SQL: GATE CSE 2009 | Question: 55

Consider the following relational schema:

$\text{Suppliers}(\underline{\text{sid:integer}}, \text{ sname:string, city:string, street:string})$

$\text{Parts}(\underline{\text{pid:integer}}, \text{ pname:string, color:string})$

$\text{Catalog}(\underline{\text{sid:integer, pid:integer}}, \text{cost:real})$

Consider the following relational query on the above database:

```
SELECT S.sname
FROM    Suppliers S
WHERE S.sid NOT IN (SELECT C.sid
                FROM Catalog C
                WHERE C.pid NOT IN (SELECT P.pid
                                FROM Parts P
                                WHERE P.color<>'blue'))
```

Assume that relations corresponding to the above schema are not empty. Which one of the following is the correct interpretation of the above query?

A. Find the names of all suppliers who have supplied a non-blue part.
B. Find the names of all suppliers who have not supplied a non-blue part.
C. Find the names of all suppliers who have supplied only non-blue part.
D. Find the names of all suppliers who have not supplied only blue parts.

gatecse-2009 databases sql normal

Answer key

3.21.25 SQL: GATE CSE 2009 | Question: 56

Consider the following relational schema:

- $\text{Suppliers}(\underline{\text{sid:integer}}, \text{ sname:string, city:string, street:string})$
- $\text{Parts}(\underline{\text{pid:integer}}, \text{ pname:string, color:string})$
- $\text{Catalog}(\underline{\text{sid:integer, pid:integer}}, \text{cost:real})$

Assume that, in the suppliers relation above, each supplier and each street within a city has unique name, and (sname, city) forms a candidate key. No other functional dependencies are implied other than those implied by primary and candidate keys. Which one of the following is TRUE about the above schema?

A. The schema is in BCNF

B. The schema is in 3NF but not in BCNF

C. The schema is in 2NF but not in 3NF

D. The schema is not in 2NF

gatecse-2009 databases sql database-normalization normal

Answer key

3.21.26 SQL: GATE CSE 2010 | Question: 19

A relational schema for a train reservation database is given below.

- **passenger(pid, pname, age)**
- **reservation(pid, class, tid)**

Passenger

pid	pname	Age
0	Sachine	65
1	Rahul	66
2	Sourav	67
3	Anil	69

Reservation

pid	class	tid
0	AC	8200
1	AC	8201
2	SC	8201
5	AC	8203
1	SC	8204
3	AC	8202

What **pids** are returned by the following SQL query for the above instance of the tables?

```
SELECT pid
FROM Reservation
WHERE class='AC' AND
   EXISTS (SELECT *
         FROM Passenger
         WHERE age>65 AND
         Passenger.pid=Reservation.pid)
```

A. $1,0$

B. $1,2$

C. $1,3$

D. $1,5$

gatecse-2010 databases sql normal

Answer key

3.21.27 SQL: GATE CSE 2011 | Question: 32

Consider a database table T containing two columns X and Y each of type integer. After the creation of the table, one record $(\text{X}{=}1, \text{Y}{=}1)$ is inserted in the table.

Let MX and MY denote the respective maximum values of X and Y among all records in the table at any point in time. Using MX and MY, new records are inserted in the table 128 times with X and Y values being $\text{MX}{+}1, 2{*}\text{MY}{+}1$ respectively. It may be noted that each time after the insertion, values of MX and MY change.

What will be the output of the following SQL query after the steps mentioned above are carried out?

```
SELECT Y FROM T WHERE X=7;
```

A. 127

B. 255

C. 129

D. 257

gatecse-2011 databases sql normal

Answer key

3.21.28 SQL: GATE CSE 2011 | Question: 46

Database table by name Loan_Records is given below.

Borrower	Bank_Manager	Loan_Amount
Ramesh	Sunderajan	10000.00
Suresh	Ramgopal	5000.00
Mahesh	Sunderajan	7000.00

What is the output of the following SQL query?

```
SELECT count(*)
FROM (
   SELECT Borrower, Bank_Manager FROM Loan_Records) AS S
   NATURAL JOIN
   (SELECT Bank_Manager, Loan_Amount FROM Loan_Records) AS T
);
```

A. 3 B. 9 C. 5 D. 6

gatecse-2011 databases sql normal

Answer key

3.21.29 SQL: GATE CSE 2012 | Question: 15

Which of the following statements are **TRUE** about an SQL query?
P : An SQL query can contain a HAVING clause even if it does not have a GROUP BY clause
Q : An SQL query can contain a HAVING clause only if it has a GROUP BY clause
R : All attributes used in the GROUP BY clause must appear in the SELECT clause
S : Not all attributes used in the GROUP BY clause need to appear in the SELECT clause

A. P and R B. P and S C. Q and R D. Q and S

gatecse-2012 databases easy sql ambiguous

Answer key

3.21.30 SQL: GATE CSE 2012 | Question: 51

Consider the following relations A, B and C :

A

Id	Name	Age
12	Arun	60
15	Shreya	24
99	Rohit	11

B

Id	Name	Age
15	Shreya	24
25	Hari	40
98	Rohit	20
99	Rohit	11

C

Id	Phone	Area
10	2200	02
99	2100	01

How many tuples does the result of the following SQL query contain?

```
SELECT A.Id
FROM A
WHERE A.Age > ALL (SELECT B.Age
              FROM B
              WHERE B.Name = 'Arun')
```

A. 4 B. 3 C. 0 D. 1

gatecse-2012 databases sql normal

Answer key

3.21.31 SQL: GATE CSE 2014 | Set 1 | Question: 22

Given the following statements:

S1: A foreign key declaration can always be replaced by an equivalent check assertion in SQL.

S2: Given the table $R(a, b, c)$ where a and b together form the primary key, the following is a valid table definition.

```
CREATE TABLE S (
    a INTEGER,
    d INTEGER,
    e INTEGER,
    PRIMARY KEY (d),
    FOREIGN KEY (a) references R)
```

Which one of the following statements is **CORRECT**?

A. S1 is TRUE and S2 is FALSE
B. Both S1 and S2 are TRUE
C. S1 is FALSE and S2 is TRUE
D. Both S1 and S2 are FALSE

gatecse-2014-set1 databases normal sql

Answer key

3.21.32 SQL: GATE CSE 2014 | Set 1 | Question: 54

Given the following schema:

employees(emp-id, first-name, last-name, hire-date, dept-id, salary)

departments(dept-id, dept-name, manager-id, location-id)

You want to display the last names and hire dates of all latest hires in their respective departments in the location $\text{ID } 1700$. You issue the following query:

```
SQL>SELECT last-name, hire-date
   FROM employees
   WHERE (dept-id, hire-date) IN
   (SELECT dept-id, MAX(hire-date)
   FROM employees JOIN departments USING(dept-id)
   WHERE location-id =1700
   GROUP BY dept-id);
```

What is the outcome?

A. It executes but does not give the correct result
B. It executes and gives the correct result.
C. It generates an error because of pairwise comparison.
D. It generates an error because of the GROUP BY clause cannot be used with table joins in a sub-query.

gatecse-2014-set1 databases sql normal

Answer key

3.21.33 SQL: GATE CSE 2014 | Set 2 | Question: 54

SQL allows duplicate tuples in relations, and correspondingly defines the multiplicity of tuples in the result of joins. Which one of the following queries always gives the same answer as the nested query shown below:

```
select * from R where a in (select S.a from S)
```

A. select R.* from R, S where R.a=S.a
B. select distinct R.* from R,S where R.a=S.a
C. select R.* from R,(select distinct a from S) as S1 where R.a=S1.a
D. select R.* from R,S where R.a=S.a and is unique R

gatecse-2014-set2 databases sql normal

Answer key

3.21.34 SQL: GATE CSE 2014 | Set 3 | Question: 54

Consider the following relational schema:

employee (empId,empName,empDept)

customer (custId,custName,salesRepId,rating)

salesRepId is a foreign key referring to **empId** of the employee relation. Assume that each employee makes a sale to at least one customer. What does the following query return?

```
SELECT empName   FROM employee E
WHERE NOT EXISTS (SELECT custId
   FROM customer C
   WHERE C.salesRepId = E.empId
   AND C.rating <> 'GOOD');
```

A. Names of all the employees with at least one of their customers having a ‘GOOD’ rating.
B. Names of all the employees with at most one of their customers having a 'GOOD' rating.
C. Names of all the employees with none of their customers having a 'GOOD' rating.
D. Names of all the employees with all their customers having a 'GOOD' rating.

gatecse-2014-set3 databases sql easy

Answer key

3.21.35 SQL: GATE CSE 2015 | Set 1 | Question: 27

Consider the following relation:

Student

Roll_No	Student_Name
1	Raj
2	Rohit
3	Raj

Performance

Roll_No	Course	Marks
1	Math	80
1	English	70
2	Math	75
3	English	80
2	Physics	65
3	Math	80

Consider the following SQL query.

```
SELECT S.Student_Name, Sum(P. Marks)
FROM Student S, Performance P
WHERE S.Roll_No= P.Roll_No
GROUP BY S.STUDENT_Name
```

The numbers of rows that will be returned by the SQL query is_________________.

gatecse-2015-set1 databases sql normal numerical-answers

Answer key

3.21.36 SQL: GATE CSE 2015 | Set 3 | Question: 3

Consider the following relation

Cinema($theater, address, capacity$)

Which of the following options will be needed at the end of the SQL query

```
SELECT P1.address
FROM Cinema P1
```

such that it always finds the addresses of theaters with maximum capacity?

A. `WHERE P1.capacity >= All (select P2.capacity from Cinema P2)`

B. `WHERE P1.capacity >= Any (select P2.capacity from Cinema P2)`

C. `WHERE P1.capacity > All (select max(P2.capacity) from Cinema P2)`

D. `WHERE P1.capacity > Any (select max(P2.capacity) from Cinema P2)`

gatecse-2015-set3 databases sql normal

Answer key

3.21.37 SQL: GATE CSE 2016 | Set 2 | Question: 52

Consider the following database table named water_schemes:

Water_schemes

scheme_no	district_name	capacity
1	Ajmer	20
1	Bikaner	10
2	Bikaner	10
3	Bikaner	20
1	Churu	10
2	Churu	20
1	Dungargarh	10

The number of tuples returned by the following SQL query is _________.

```
with total (name, capacity) as
    select district_name, sum (capacity)
    from water_schemes
    group by district_name
with total_avg (capacity) as
   select avg (capacity)
   from total
select name
   from total, total_avg
   where total.capacity ≥ total_avg.capacity
```

gatecse-2016-set2 databases sql normal numerical-answers

Answer key

3.21.38 SQL: GATE CSE 2017 | Set 1 | Question: 23

Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName). An instance of the schema EMP and a SQL query on it are given below:

EMP

EmpId	EmpName	DeptName
1	XYA	AA
2	XYB	AA
3	XYC	AA
4	XYD	AA
5	XYE	AB
6	XYF	AB
7	XYG	AB
8	XYH	AC
9	XYI	AC
10	XYJ	AC
11	XYK	AD
12	XYL	AD
13	XYM	AE

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
    (SELECT DeptName, COUNT(EmpId) AS
                    EC(DeptName, Num)
    FROM EMP
    GROUP BY DeptName)
```

The output of executing the SQL query is _____________ .